STAT3888: Post-Feedback Revisions

AUTHOR
Calvin Lai

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The post-feedback revisions will include the following changes:

- Replacing the categorical `WAIST` with `PHDCMWBC` for continuous waist circumference.
- Reform the alcohol consumption level and smoking duration as confounders instead of filtering variables, and remove `DSMKHHRB`.
- Change `AGEB` to numerical values by taking mid-points, to hopefully increase power and reduce number of parameters.
- Adding income (`INCDECPN`) and education level (`EDATTQ2`) as more confounders as well for socioeconomic status proxies.
- After fitting the multinomial logistic regression model under these new variables, maybe try converting ALT and GGT to continuous variables and fitting a regression model.

Dataset Rework

▼ Code

```
# Initialisation
knitr::opts_chunk$set(warning = FALSE, message = FALSE,
                      fig.align='center', fig.dim=c(10, 8))

library(tidyverse)
library(here)
library(naniar)
library(nnet)
library(broom)
library(knitr)
library(car)
library(gglm)
library(patchwork)
library(effects)
library(mice)
library(mgcv)
library(MASS)
library(brant)

# Take original PHDCMWBC from BSP
bsp.raw = read.csv(here("NHS/AHSnhs11bsp.csv"))
```

```
waist.raw = bsp.raw$PHDCMWBC
waist.cont = case_when(waist.raw %in% c(0, 998, 999) ~ NA,
                       TRUE ~ waist.raw)

# New age variable
age.raw = bsp.raw$AGEB
age.vals = case_when(age.raw %in% c(1, 2, 3, 4) ~ NA,
                age.raw == 5 ~ 18.5,
                age.raw == 6 ~ 22,
                age.raw == 7 ~ 27,
                age.raw == 8 ~ 32,
                age.raw == 9 ~ 37,
                age.raw == 10 ~ 42,
                age.raw == 11 ~ 47,
                age.raw == 12 ~ 52,
                age.raw == 13 ~ 57,
                age.raw == 14 ~ 62,
                age.raw == 15 ~ 67,
                age.raw == 16 ~ 72,
                age.raw == 17 ~ 77,
                age.raw == 18 ~ 82,
                age.raw == 19 ~ 87)
age = age.vals

# Variable lists
bbi.vars = c("ABSLID", "ABSPID", "ALTRESB", "GGTRESB")
bsp.vars = c("ABSLID", "ABSPID", "AL2K7DAY", "SMKAGEDB", "EXREGU
             "SEX", "EDATTQ2", "INCDECPN")

# Load data sets
bsp = readRDS(here("Objects", "AHSnhs11bsp.rds"))
bbi = readRDS(here("Objects", "AHSnhs11bbi.rds"))

# Variable tables
bbir = bbi %>% dplyr::select(all_of(bbi.vars)) %>%
  mutate(ALTRESB = fct_relevel(ALTRESB, levels(bbi$ALTRESB)[9]))
bspr = bsp %>% dplyr::select(all_of(bsp.vars)) %>%
  mutate(EXREGUIC = fct_relevel(EXREGUIC, levels(bsp$EXREGUIC)[2
         AGE = age,
         WAIST = waist.cont)

# Merge data tables and original ALT3 and GGT3 for
dat = inner_join(bspr, bbir, by = c("ABSLID", "ABSPID")) %>% dis
dat = dat %>%
  mutate(
    ALT3 = case_when(
      SEX == "Male" & ALTRESB %in% c("Less than 10",
                                     "10 to less than 15",
```

```
                                    "15 to less than 20",
                                    "20 to less than 25",
                                    "25 to less than 30") ~ "Lo
      SEX == "Male" & ALTRESB %in% c("30 to less than 35",
                                    "35 to less than 40") ~ "Bo
      SEX == "Male" & ALTRESB %in% c("40 to less than 45",
                                    "45 to less than 50",
                                    "More than 50") ~ "High",

      SEX == "Female" & ALTRESB %in% c("Less than 10",
                                    "10 to less than 15",
                                    "15 to less than 20",
                                    "20 to less than 25") ~ "Lo
      SEX == "Female" & ALTRESB == "25 to less than 30" ~ "Borde
      SEX == "Female" & ALTRESB %in% c("30 to less than 35",
                                    "35 to less than 40",
                                    "40 to less than 45",
                                    "45 to less than 50",
                                    "More than 50") ~ "High",
      TRUE ~ NA
    ) %>% as.factor(),
    GGT3 = case_when(
      SEX == "Male" & GGTRESB %in% c("10 or less", "More than 10
                                    "More than 15 to 20",
                                    "More than 20 to 25",
                                    "More than 25 to 30",
                                    "More than 30 to 35",
                                    "More than 35 to 40",
                                    "More than 40 to 45") ~ "Lo
      SEX == "Male" & GGTRESB %in% c("More than 45 to 50",
                                    "More than 50 to 55")  ~ "M
      SEX == "Male" & GGTRESB %in% c("More than 55 to 60",
                                    "More than 60") ~ "High",

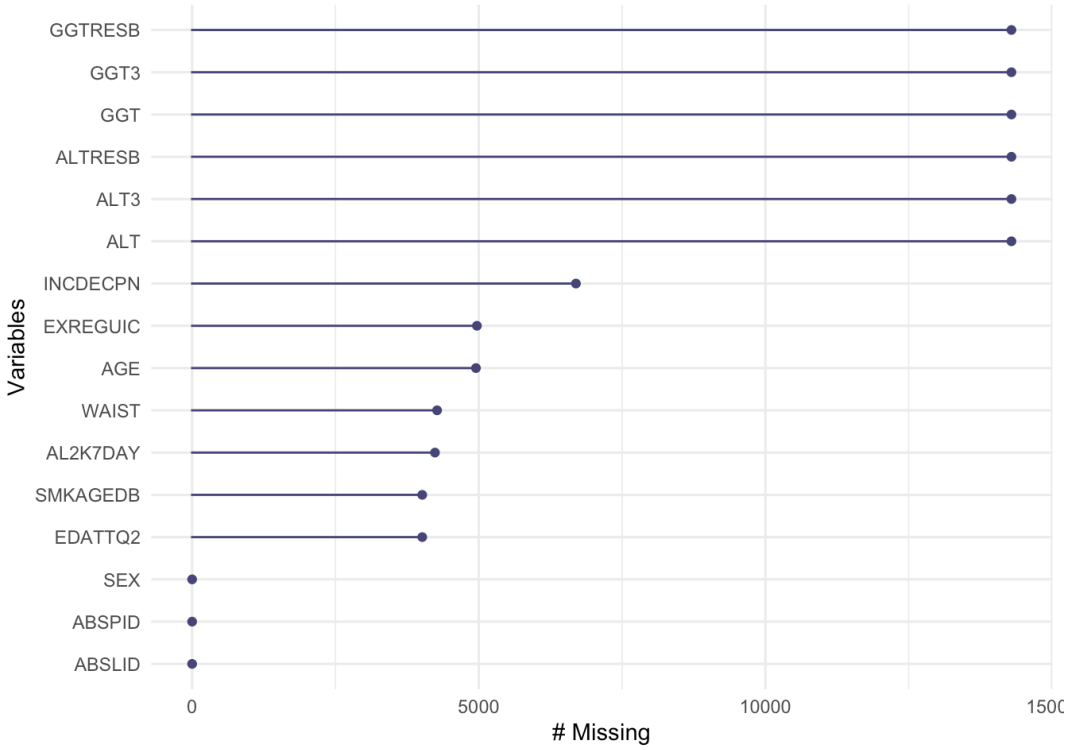
      SEX == "Female" & GGTRESB %in% c("10 or less", "More than
                                    "More than 15 to 20",
                                    "More than 20 to 25",
                                    "More than 25 to 30") ~ "Lo
      SEX == "Female" & GGTRESB == "More than 30 to 35" ~ "Borde
      SEX == "Female" & GGTRESB %in% c("More than 35 to 40",
                                    "More than 40 to 45",
                                    "More than 45 to 50",
                                    "More than 50 to 55",
                                    "More than 55 to 60",
                                    "More than 60") ~ "High",
      TRUE ~ NA
    ) %>% as.factor(),
    EXREGUIC = relevel(EXREGUIC, ref = "Met recommended guidelin
```

```
  ) %>%
  mutate(ALT3 = fct_relevel(ALT3, "Low", "Borderline", "High"),
         GGT3 = fct_relevel(GGT3, "Low", "Borderline", "High"))

# Make numerical ALT and GGT
set.seed(3888)
alt.raw = dat$ALTRESB
alt.vals = case_when(
  alt.raw == "Less than 10" ~ 5,
  alt.raw == "10 to less than 15" ~ 12.5,
  alt.raw == "15 to less than 20" ~ 17.5,
  alt.raw == "20 to less than 25" ~ 22.5,
  alt.raw == "25 to less than 30" ~ 27.5,
  alt.raw == "30 to less than 35" ~ 32.5,
  alt.raw == "35 to less than 40" ~ 37.5,
  alt.raw == "40 to less than 45" ~ 42.5,
  alt.raw == "45 to less than 50" ~ 47.5,
  alt.raw == "More than 50" ~ 52.5)
ALT = alt.vals

set.seed(3882)
ggt.raw = dat$GGTRESB
ggt.vals = case_when(
  ggt.raw == "10 or less" ~ 5,
  ggt.raw == "More than 10 to 15" ~ 12.5,
  ggt.raw == "More than 15 to 20" ~ 17.5,
  ggt.raw == "More than 20 to 25" ~ 22.5,
  ggt.raw == "More than 25 to 30" ~ 27.5,
  ggt.raw == "More than 30 to 35" ~ 32.5,
  ggt.raw == "More than 35 to 40" ~ 37.5,
  ggt.raw == "More than 40 to 45" ~ 42.5,
  ggt.raw == "More than 45 to 50" ~ 47.5,
  ggt.raw == "More than 50 to 55" ~ 52.5,
  ggt.raw == "More than 55 to 60" ~ 57.5,
  ggt.raw == "More than 60" ~ 62.5)
GGT = ggt.vals

# Complete data
dat = dat %>% mutate(ALT = ALT, GGT = GGT)
gg_miss_var(dat)
```



▼ Code

```
miss_var_summary(dat)
```

```
# A tibble: 16 × 3
   variable n_miss pct_miss
   <chr>     <int>    <num>
 1 ALTRESB   14291     70.0
 2 GGTRESB   14291     70.0
 3 ALT3      14291     70.0
 4 GGT3      14291     70.0
 5 ALT       14291     70.0
 6 GGT       14291     70.0
 7 INCDECPN   6695     32.8
 8 EXREGUIC   4968     24.3
 9 AGE        4951     24.2
10 WAIST      4273     20.9
11 AL2K7DAY   4236     20.7
12 SMKAGEDB   4014     19.7
13 EDATTQ2    4014     19.7
14 ABSLID        0      0
15 ABSPID        0      0
16 SEX           0      0
```

▼ Code

```
# Remove missing data, and cleaning categories
```

```
dat.full = dat %>% drop_na() %>% mutate(
  AL2K7DAY = case_when(AL2K7DAY %in% c("Medium risk", "High risk
                       ~ "Medium/High risk",
                       .default = "Low risk") %>% as.factor(),
  SMKAGEDB = fct_relevel(SMKAGEDB, "Never smoked daily", "Less t
                         "10 to 19 years", "20 to 29 years", "30
                         "40 to 49 years", "50 years and over"),
  EXREGUIC = fct_recode(EXREGUIC,
                        "Yes" = "Met recommended guidelines",
                        "No" = "Did not meet recommended guideli
  EDATTQ2 = fct_recode(EDATTQ2,
                       "No" = "Does not have non-school qualific
                       "Yes" = "Has non-school qualification"),
  ) %>% rename(
    "AlcoholRisk" = AL2K7DAY,
    "SmokingDuration" = SMKAGEDB,
    "ExerciseGuidelines" = EXREGUIC,
    "NonSchoolQuals" = EDATTQ2,
    "Sex" = SEX,
    "Age" = AGE,
    "IncomeDecile" = INCDECPN,
    "Waist" = WAIST,
    "ALT10" = ALTRESB,
    "GGT12" = GGTRESB
  )
```

In this case, there are now **4694** complete cases, which I think should be enough for our analysis. We list out the variables used as well using `names()` as a check, and check each variable numerically.

▼ Code

```
# Check variables
names(dat.full)
```

```
 [1] "ABSLID"             "ABSPID"             "AlcoholRisk"
 [4] "SmokingDuration"    "ExerciseGuidelines" "Sex"
 [7] "NonSchoolQuals"     "IncomeDecile"       "Age"
[10] "Waist"              "ALT10"              "GGT12"
[13] "ALT3"               "GGT3"               "ALT"
[16] "GGT"
```

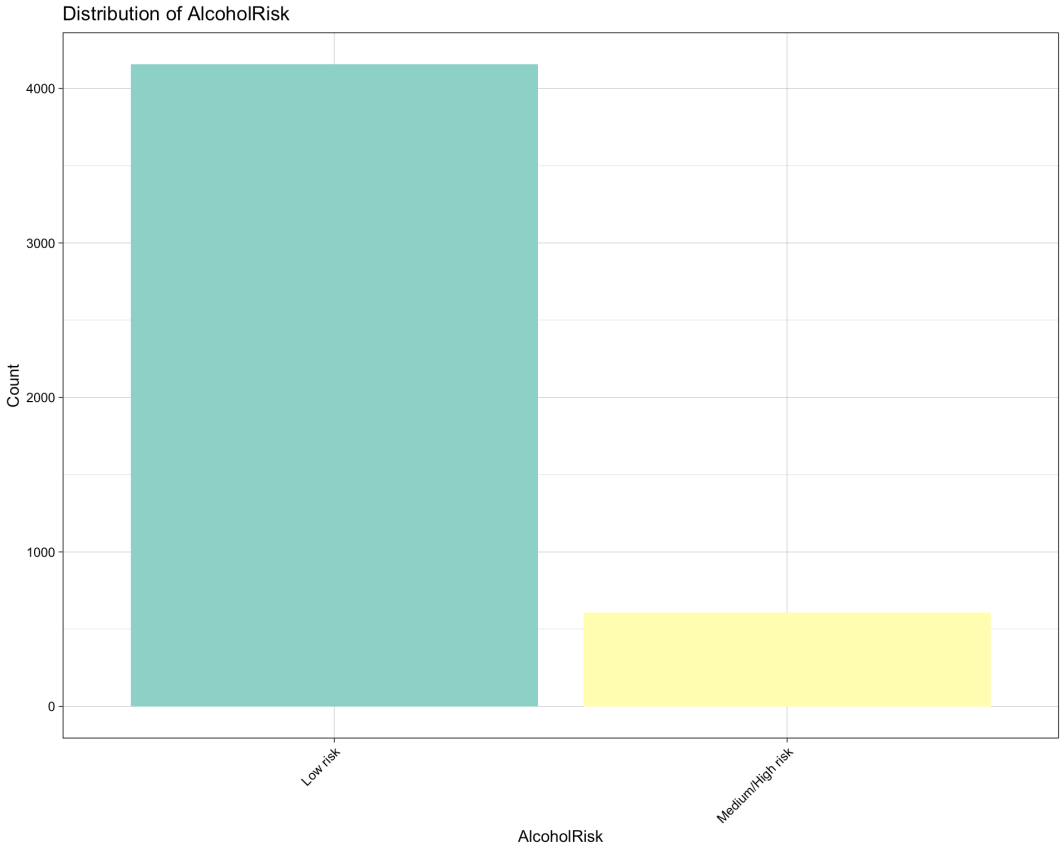
▼ Code

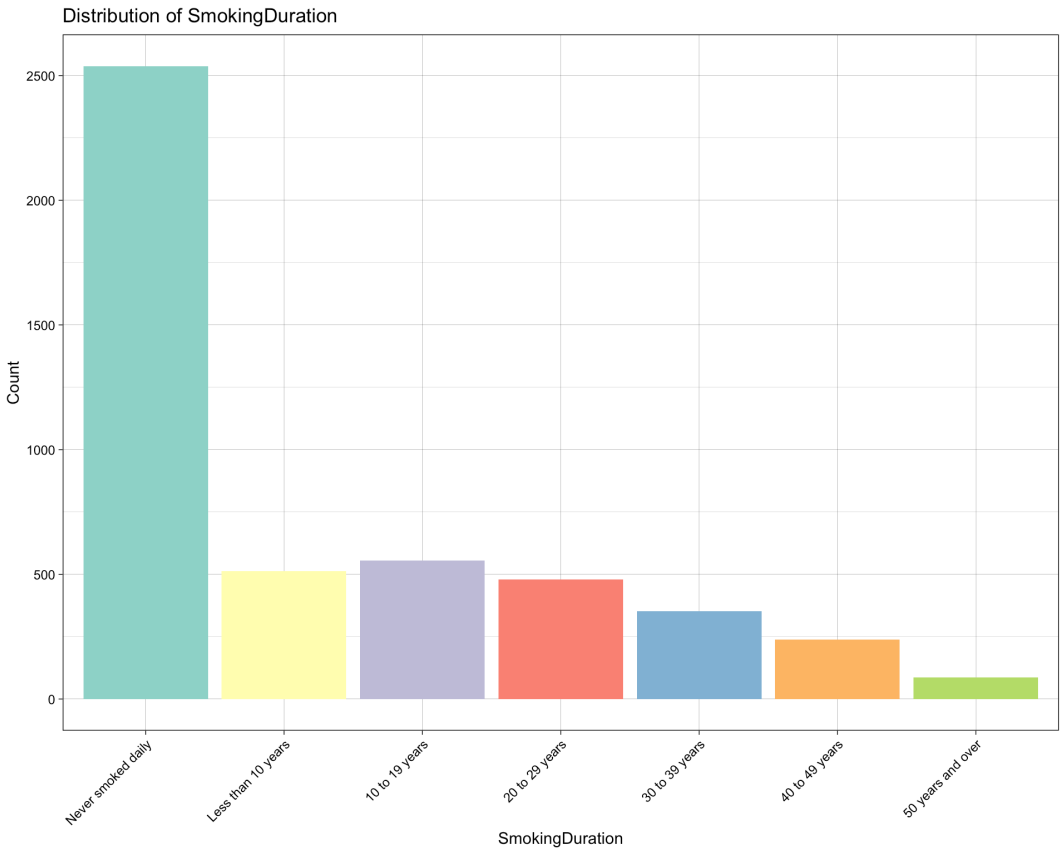
```
# Categorical variables
categoricals = c("AlcoholRisk", "SmokingDuration", "ExerciseGuid
```

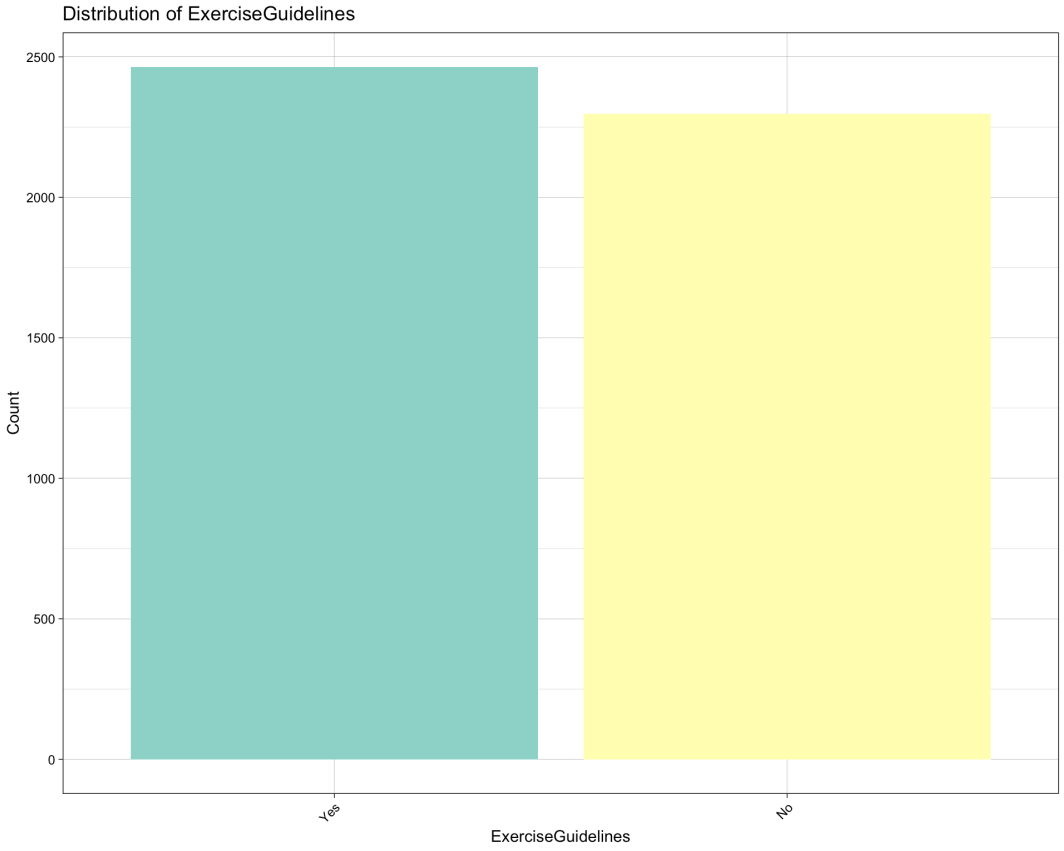
```
                  "NonSchoolQuals", "IncomeDecile", "ALT10", "GG
for (var in categoricals) {

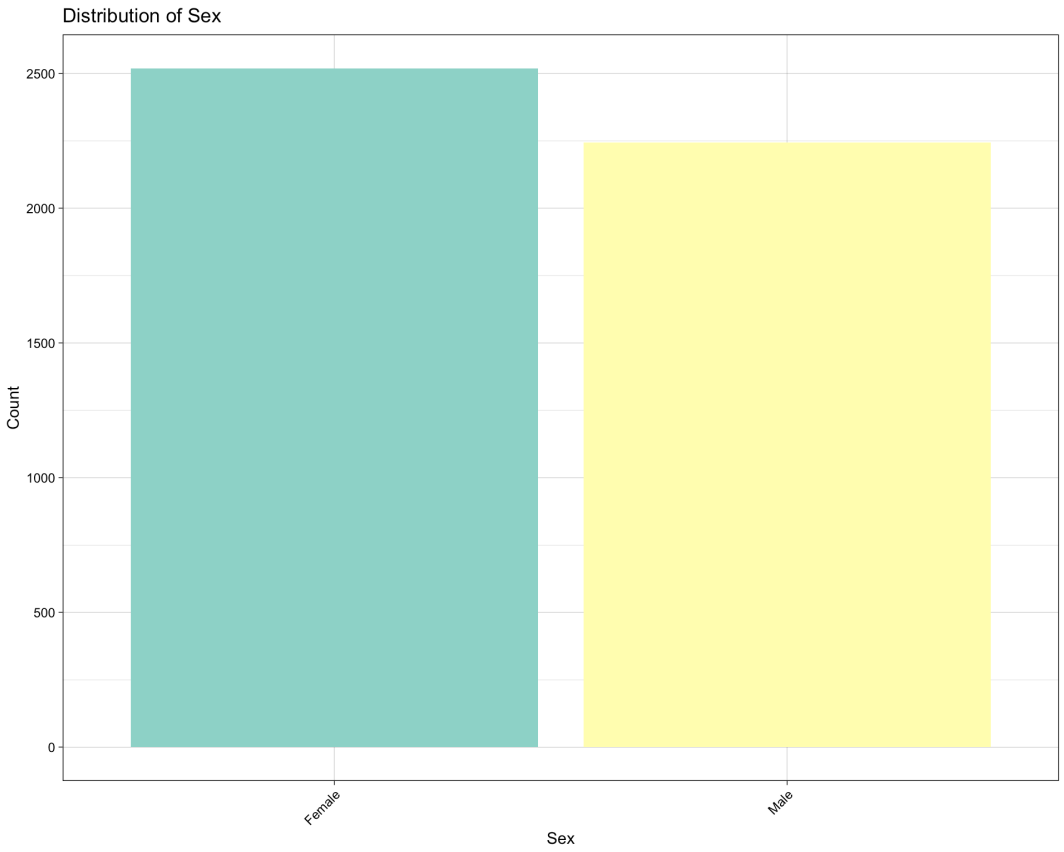
  p = ggplot(dat.full, aes_string(x = var, fill = var)) +
    geom_bar() +
    labs(title = paste("Distribution of", var), x = var, y = "C
    scale_fill_brewer(palette = "Set3") +
    theme_linedraw() +
    theme(axis.text.x = element_text(angle = 45, hjust = 1),
          legend.position = "none")
  print(p)

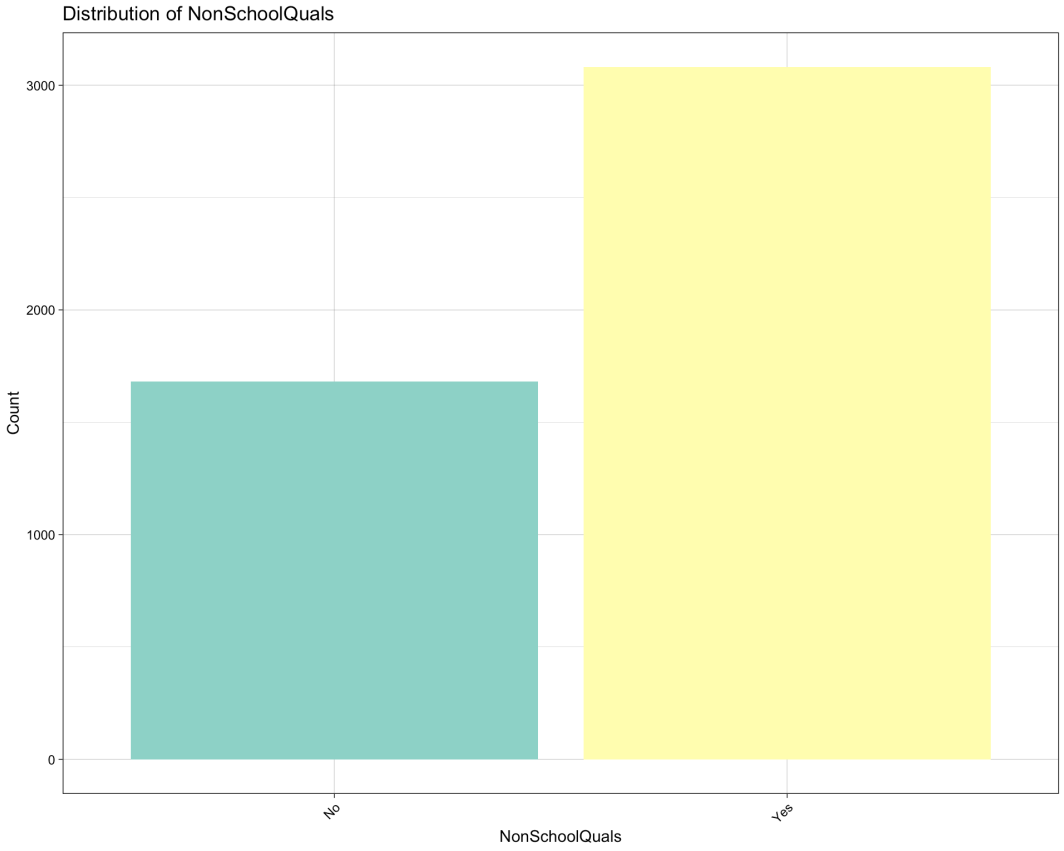
}
```

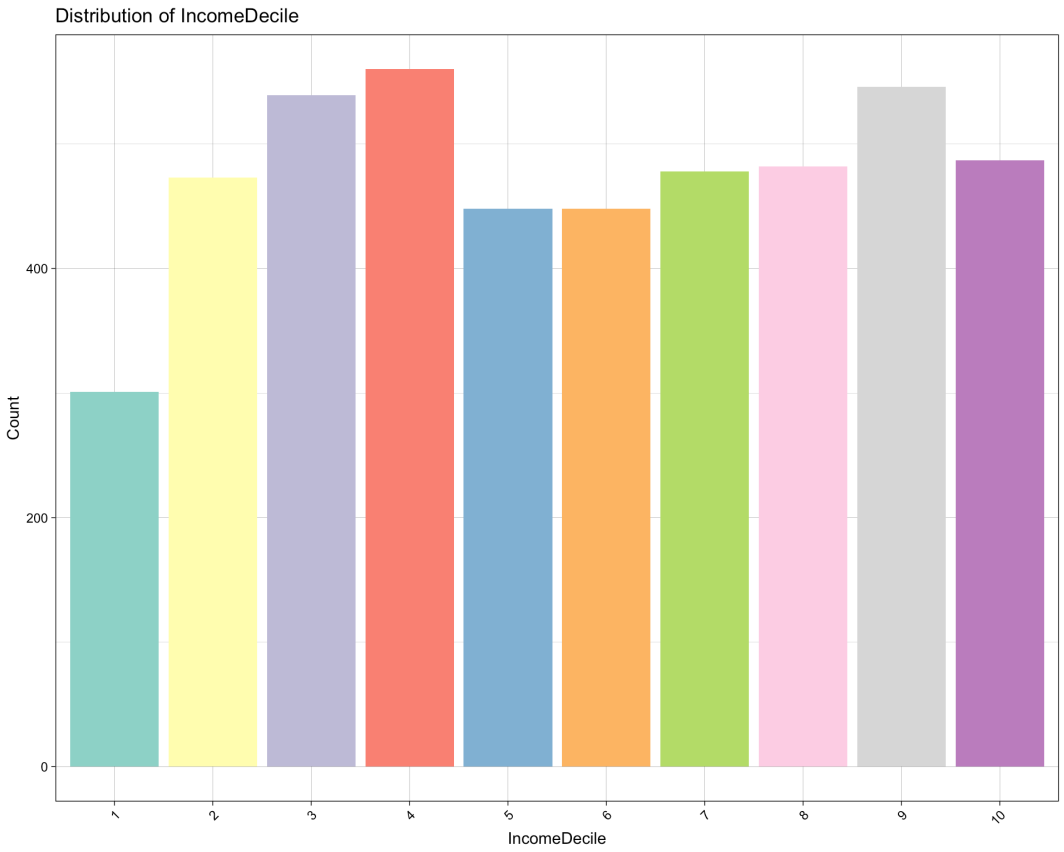


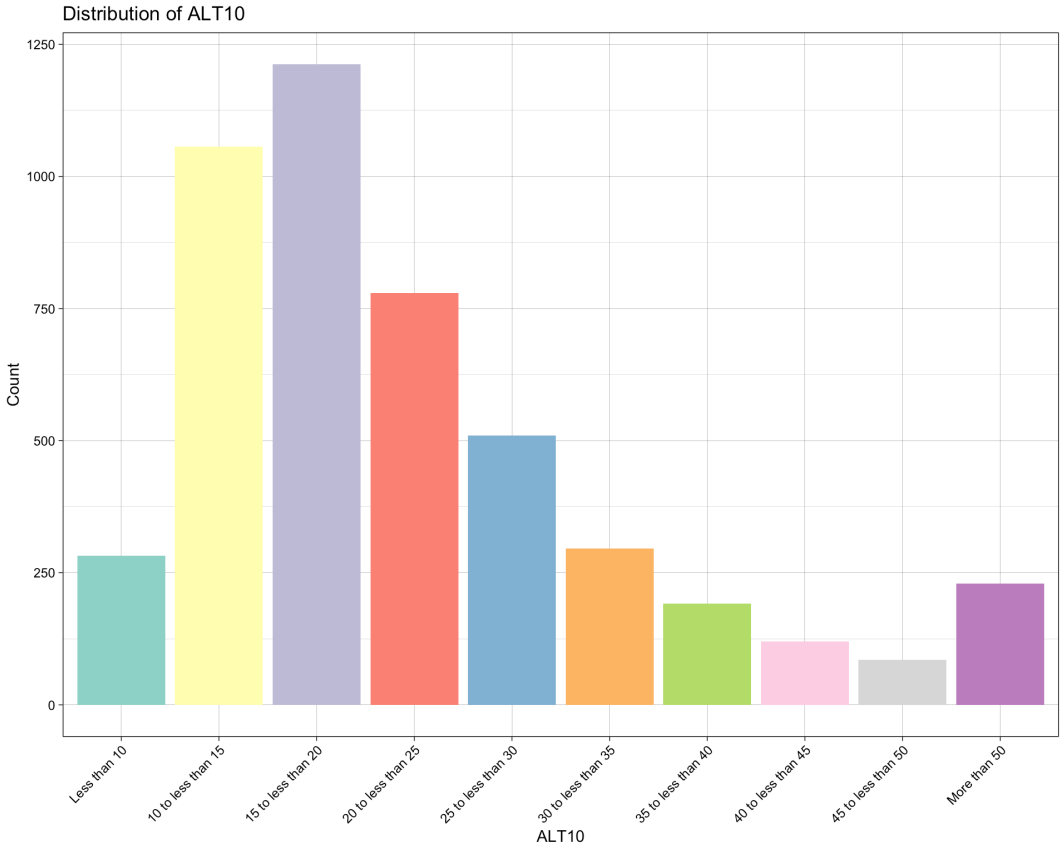

Distribution of SmokingDuration
Count
2500
2000
1500
1000
500
0
Never smoked daily
Less than 10 years
10 to 19 years
20 to 29 years
30 to 39 years
40 to 49 years
50 years and over
SmokingDuration

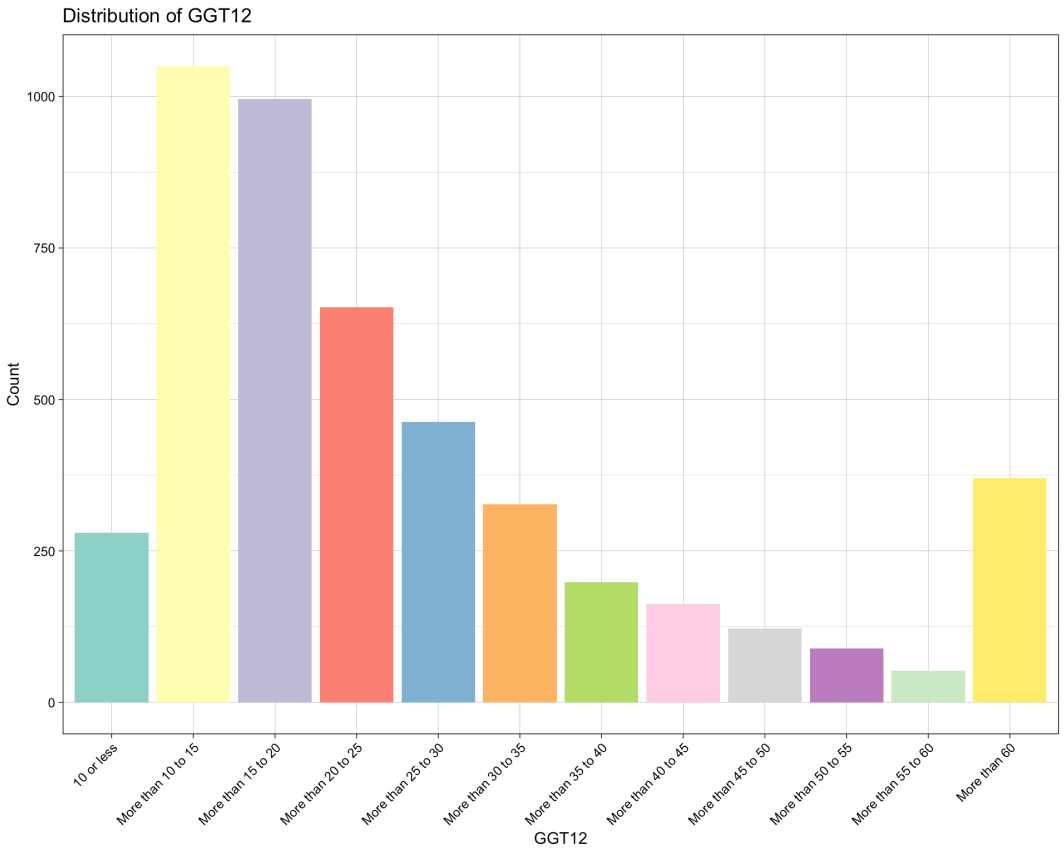

Distribution of ExerciseGuidelines
Count
2500
2000
1500
1000
500
0
Yes
No
ExerciseGuidelines

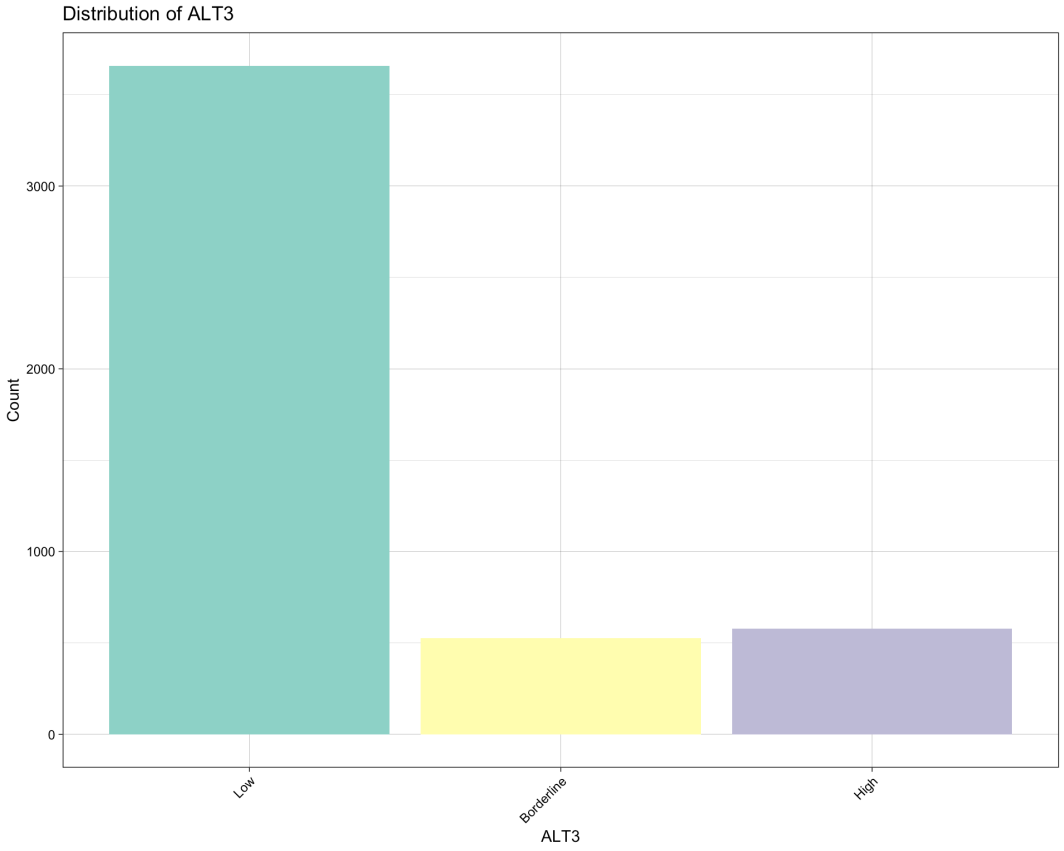

Distribution of Sex
Count
2500
2000
1500
1000
500
0
Female
Male
Sex

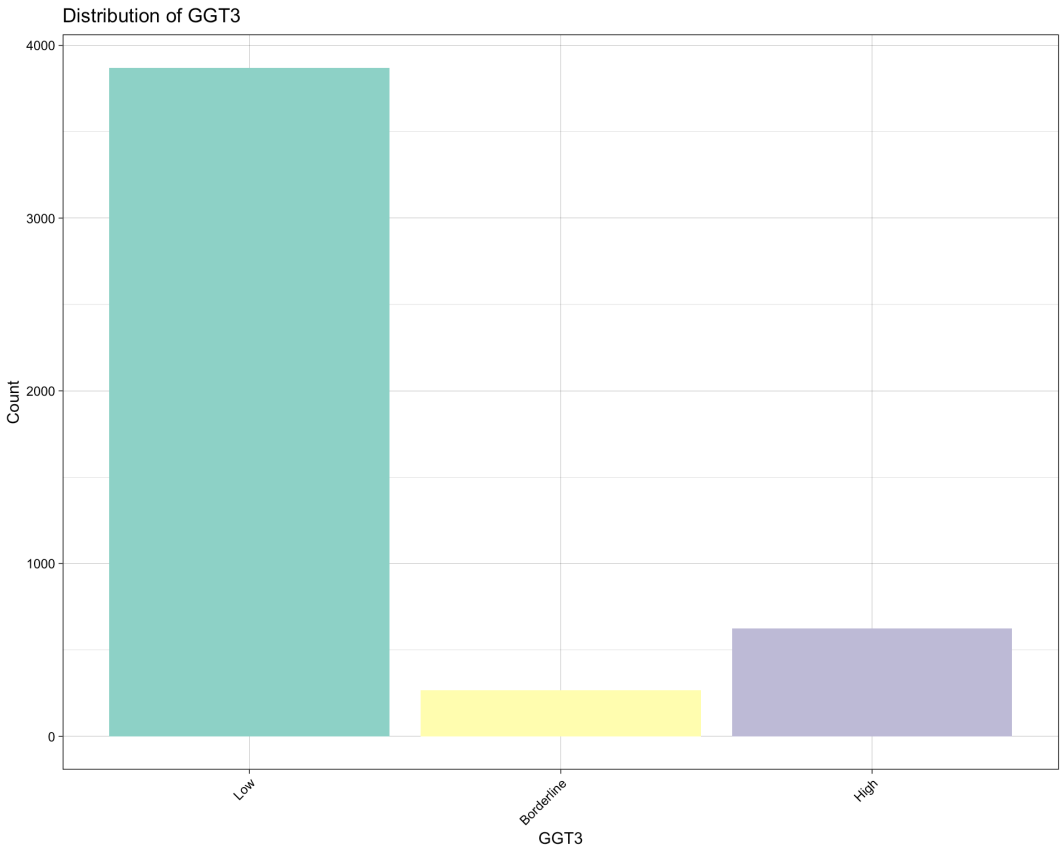

Distribution of NonSchoolQuals
Count
3000
2000
1000
0
No
Yes
NonSchoolQuals


Distribution of IncomeDecile
Count
0
200
400
1
2
3
4
5
6
7
8
9
10
IncomeDecile


Distribution of ALT10
Count
0
250
500
750
1000
1250
Less than 10
10 to less than 15
15 to less than 20
20 to less than 25
25 to less than 30
30 to less than 35
35 to less than 40
40 to less than 45
45 to less than 50
More than 50
ALT10

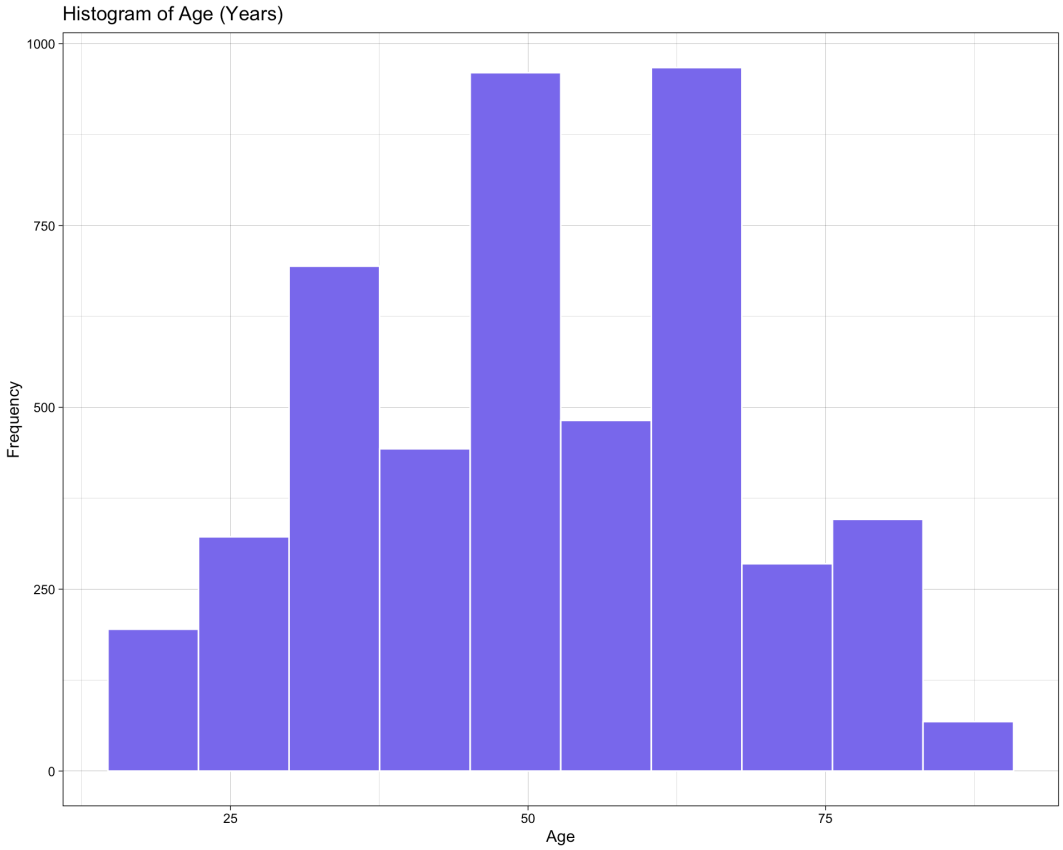

Distribution of GGT12
Count
1000
750
500
250
0
10 or less
More than 10 to 15
More than 15 to 20
More than 20 to 25
More than 25 to 30
More than 30 to 35
More than 35 to 40
More than 40 to 45
More than 45 to 50
More than 50 to 55
More than 55 to 60
More than 60
GGT12


Distribution of ALT3
Count
3000
2000
1000
0
Low
Borderline
High
ALT3



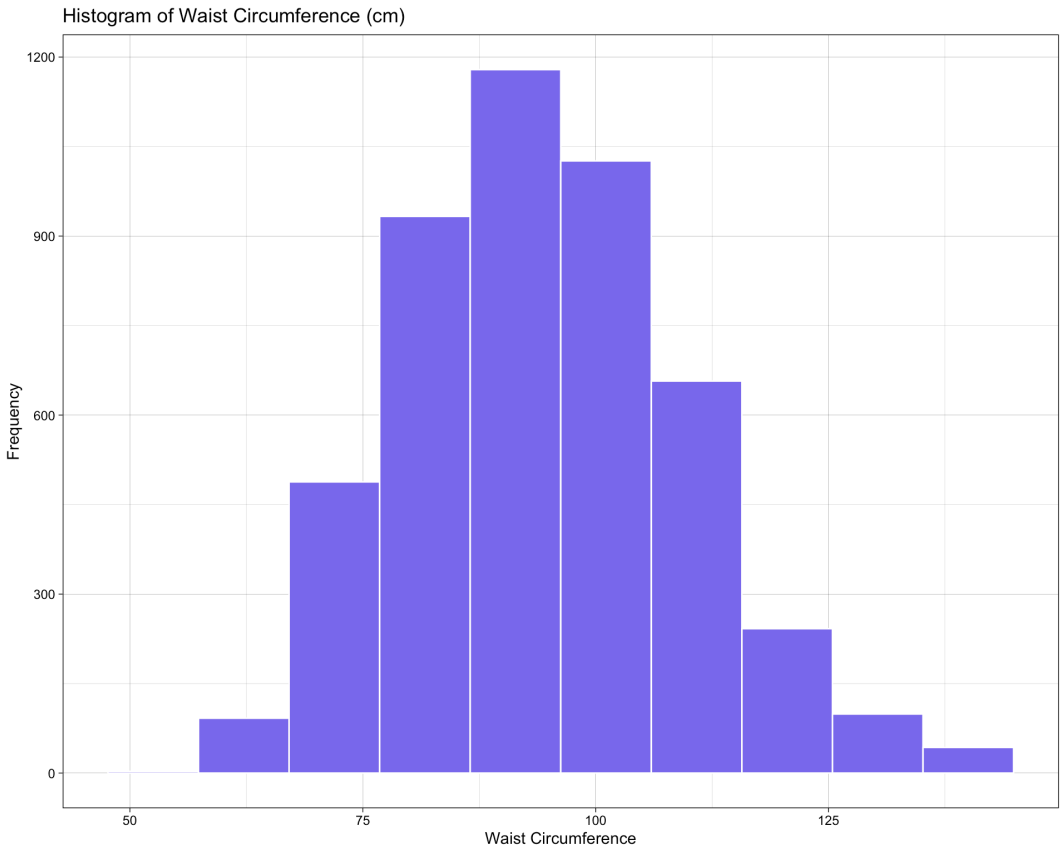
▼ Code

```
# Numeric variables
ggplot(data = dat.full, aes(x = Age)) +
  geom_histogram(bins = 10, fill = "mediumslateblue", color = "
  labs(title = "Histogram of Age (Years)", x = "Age", y = "Frequ
  theme_linedraw()
```



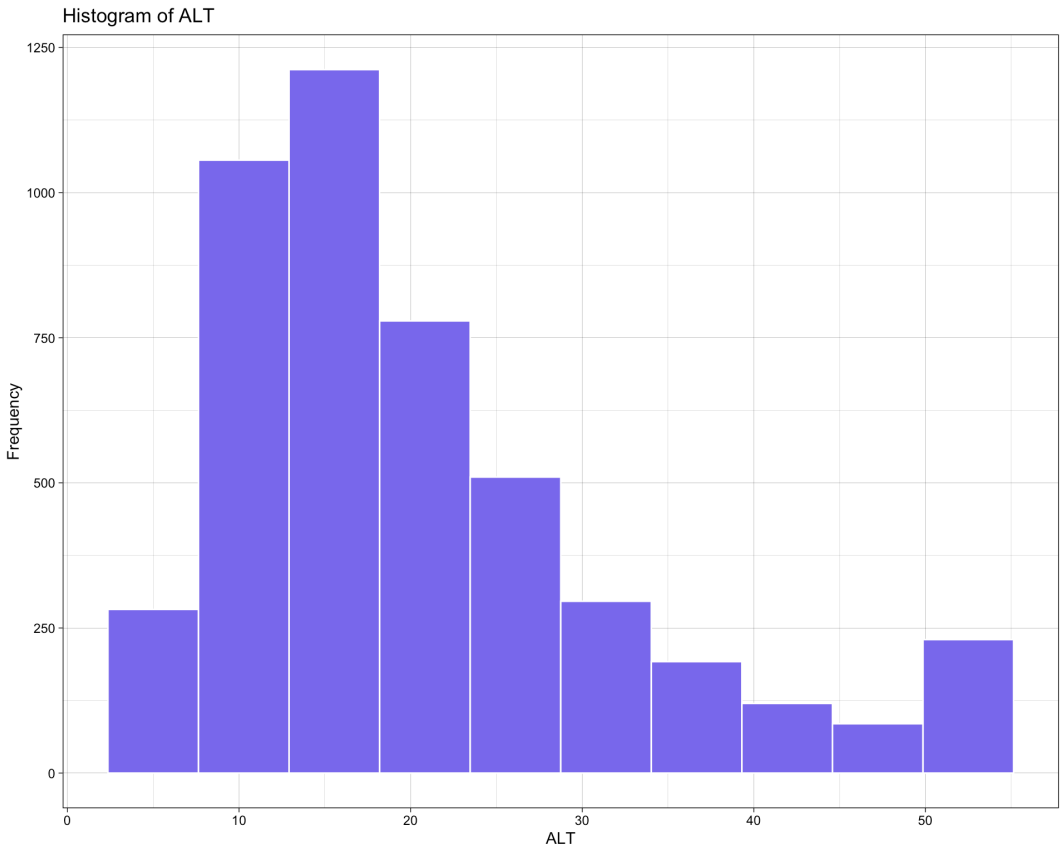
▼ Code

```
ggplot(data = dat.full, aes(x = Waist)) +
  geom_histogram(bins = 10, fill = "mediumslateblue", color = "
  labs(title = "Histogram of Waist Circumference (cm)",
       x = "Waist Circumference", y = "Frequency") +
  theme_linedraw()
```



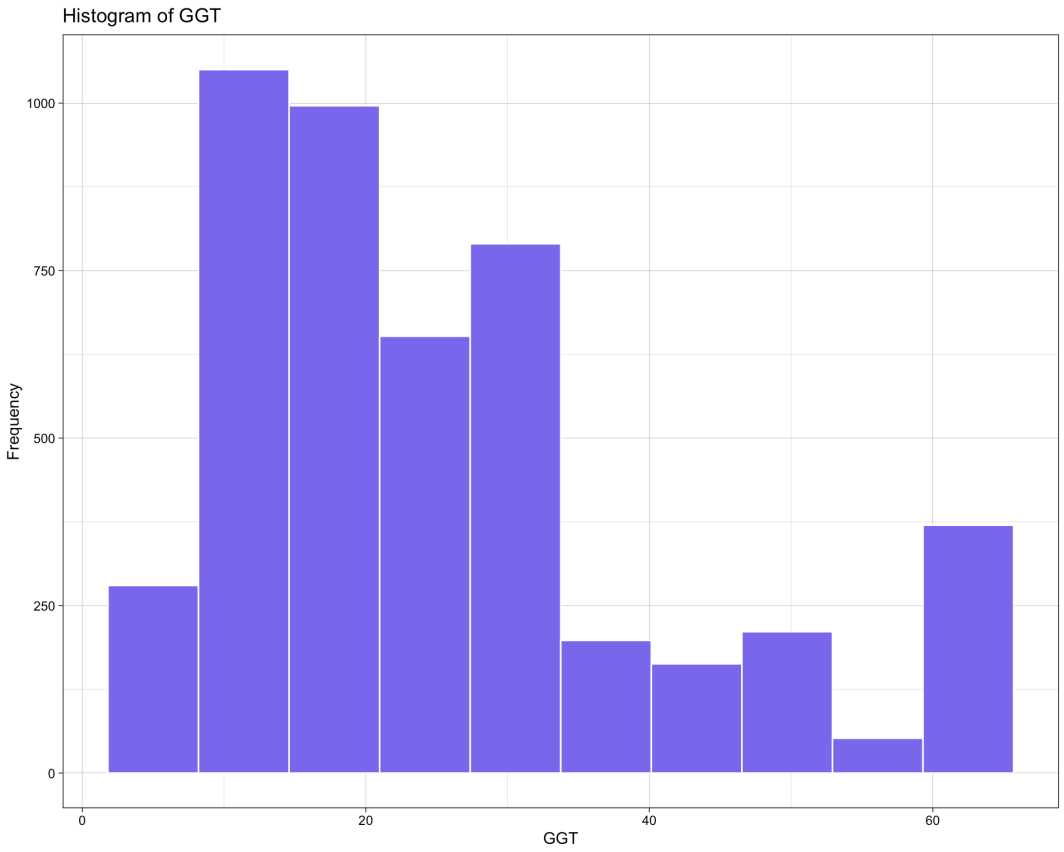
▼ Code

```
ggplot(data = dat.full, aes(x = ALT)) +
  geom_histogram(bins = 10, fill = "mediumslateblue", color = "
  labs(title = "Histogram of ALT", x = "ALT", y = "Frequency") +
  theme_linedraw()
```



▼ Code

```
ggplot(data = dat.full, aes(x = GGT)) +
  geom_histogram(bins = 10, fill = "mediumslateblue", color = "
  labs(title = "Histogram of GGT", x = "GGT", y = "Frequency") +
  theme_linedraw()
```



Multinomial Logistic Regression Fitting

We fit using the same `nnet::multinom()` function as before, for `ALT3` and `GGT3`, and then also a regular linear model for `ALT` and `GGT`.

▼ Code

```
# Multinomial Logistic Regression
ALT.multinom = nnet::multinom(ALT3 ~ ExerciseGuidelines + Sex +
                              SmokingDuration + NonSchoolQuals + In
                              Age + Waist, data = dat.full, Hess =
```

```
# weights:  69 (44 variable)
initial  value 5231.591719
iter  10 value 3332.778645
iter  20 value 3113.516122
iter  30 value 3083.159558
iter  40 value 3082.606611
final  value 3082.585935
converged
```

▼ Code

```
tidy(ALT.multinom) %>%
  kable(caption = "ALT3 Multinomial Logistic Regression Summary
        digits = 4, align = "cccccc")
```

ALT3 Multinomial Logistic Regression Summary Table

y.level	term	estimate	std.error	statistic	p.value
Borderline	(Intercept)	-5.2710	0.4082	-12.9134	0.0000
Borderline	ExerciseGuidelinesNo	-0.1195	0.0993	-1.2031	0.2289
Borderline	SexMale	0.5768	0.1069	5.3936	0.0000
Borderline	AlcoholRiskMedium/High risk	0.1229	0.1428	0.8606	0.3895
Borderline	SmokingDurationLess than 10 years	0.3287	0.1480	2.2202	0.0264
Borderline	SmokingDuration10 to 19 years	0.1409	0.1520	0.9268	0.3540
Borderline	SmokingDuration20 to 29 years	0.0097	0.1661	0.0582	0.9536
Borderline	SmokingDuration30 to 39 years	-0.0398	0.2009	-0.1983	0.8428
Borderline	SmokingDuration40 to 49 years	0.0814	0.2359	0.3452	0.7299
Borderline	SmokingDuration50 years and over	-0.5268	0.4470	-1.1786	0.2386
Borderline	NonSchoolQualsYes	0.1117	0.1092	1.0228	0.3064
Borderline	IncomeDecile2	-0.2311	0.2633	-0.8777	0.3801
Borderline	IncomeDecile3	-0.7575	0.2896	-2.6157	0.0089
Borderline	IncomeDecile4	0.0896	0.2494	0.3591	0.7196
Borderline	IncomeDecile5	-0.1170	0.2647	-0.4419	0.6585
Borderline	IncomeDecile6	0.1801	0.2535	0.7106	0.4773
Borderline	IncomeDecile7	-0.1297	0.2569	-0.5050	0.6136
Borderline	IncomeDecile8	0.1503	0.2497	0.6020	0.5472
Borderline	IncomeDecile9	0.2400	0.2428	0.9887	0.3228
Borderline	IncomeDecile10	0.1782	0.2454	0.7263	0.4677
Borderline	Age	-0.0069	0.0035	-1.9870	0.0469
Borderline	Waist	0.0344	0.0036	9.5665	0.0000
High	(Intercept)	-5.8319	0.3813	-15.2947	0.0000
High	ExerciseGuidelinesNo	0.1059	0.0966	1.0969	0.2727
High	SexMale	0.2529	0.1007	2.5112	0.0120
High	AlcoholRiskMedium/High risk	0.3894	0.1320	2.9503	0.0032

High	SmokingDurationLess than 10 years	0.0391	0.1605	0.2436	0.8075
High	SmokingDuration10 to 19 years	0.2852	0.1445	1.9740	0.0484
High	SmokingDuration20 to 29 years	0.2007	0.1561	1.2861	0.1984
High	SmokingDuration30 to 39 years	0.3753	0.1741	2.1565	0.0310
High	SmokingDuration40 to 49 years	0.4173	0.2102	1.9852	0.0471
High	SmokingDuration50 years and over	-0.5645	0.4510	-1.2517	0.2107
High	NonSchoolQualsYes	0.1861	0.1050	1.7723	0.0763
High	IncomeDecile2	-0.5033	0.2391	-2.1050	0.0353
High	IncomeDecile3	-0.2344	0.2299	-1.0194	0.3080
High	IncomeDecile4	-0.3676	0.2342	-1.5696	0.1165
High	IncomeDecile5	-0.1589	0.2296	-0.6920	0.4890
High	IncomeDecile6	-0.3133	0.2356	-1.3295	0.1837
High	IncomeDecile7	-0.4406	0.2312	-1.9054	0.0567
High	IncomeDecile8	-0.2116	0.2282	-0.9272	0.3538
High	IncomeDecile9	-0.3228	0.2265	-1.4251	0.1541
High	IncomeDecile10	-0.4723	0.2356	-2.0044	0.0450
High	Age	-0.0227	0.0035	-6.5536	0.0000
High	Waist	0.0512	0.0034	15.2724	0.0000

It seems that the effect of `ExerciseGuidelines` is not significant in either level of `ALT3`. We check for `GGT3` as well.

▼ Code

```
# Multinomial Logistic Regression
GGT.multinom = nnet::multinom(GGT3 ~ ExerciseGuidelines + Sex +
                              SmokingDuration + NonSchoolQuals + In
                              Age + Waist, data = dat.full, Hess =
```

```
# weights:  69 (44 variable)
initial  value 5231.591719
iter  10 value 3091.392369
iter  20 value 2669.879096
iter  30 value 2635.855949
iter  40 value 2634.622471
iter  50 value 2634.533708
final  value 2634.532474
converged
```

▼ Code

```
tidy(GGT.multinom) %>%
  kable(caption = "GGT3 Multinomial Logistic Regression Summary
        digits = 4, align = "cccccc")
```

GGT3 Multinomial Logistic Regression Summary Table

y.level	term	estimate	std.error	statistic	p.value
Borderline	(Intercept)	-5.5956	0.5418	-10.3273	0.0000
Borderline	ExerciseGuidelinesNo	0.1258	0.1320	0.9524	0.3409
Borderline	SexMale	-0.0691	0.1408	-0.4910	0.6234
Borderline	AlcoholRiskMedium/High risk	0.6660	0.1681	3.9620	0.0001
Borderline	SmokingDurationLess than 10 years	0.4397	0.2022	2.1747	0.0297
Borderline	SmokingDuration10 to 19 years	0.5110	0.1895	2.6963	0.0070
Borderline	SmokingDuration20 to 29 years	0.2933	0.2159	1.3582	0.1744
Borderline	SmokingDuration30 to 39 years	0.0920	0.2643	0.3480	0.7279
Borderline	SmokingDuration40 to 49 years	0.2820	0.2974	0.9480	0.3431
Borderline	SmokingDuration50 years and over	1.0171	0.3672	2.7701	0.0056
Borderline	NonSchoolQualsYes	-0.0561	0.1416	-0.3960	0.6921
Borderline	IncomeDecile2	-0.0164	0.3456	-0.0474	0.9622
Borderline	IncomeDecile3	0.1307	0.3339	0.3913	0.6956
Borderline	IncomeDecile4	-0.1641	0.3467	-0.4734	0.6359
Borderline	IncomeDecile5	0.0830	0.3432	0.2420	0.8088
Borderline	IncomeDecile6	-0.4899	0.3842	-1.2753	0.2022
Borderline	IncomeDecile7	0.0949	0.3377	0.2811	0.7786
Borderline	IncomeDecile8	0.2242	0.3356	0.6679	0.5042
Borderline	IncomeDecile9	0.0164	0.3387	0.0484	0.9614
Borderline	IncomeDecile10	-0.0154	0.3503	-0.0439	0.9650
Borderline	Age	0.0007	0.0047	0.1535	0.8780
Borderline	Waist	0.0270	0.0046	5.8990	0.0000
High	(Intercept)	-6.0581	0.3828	-15.8245	0.0000
High	ExerciseGuidelinesNo	0.2999	0.0939	3.1959	0.0014
High	SexMale	-0.5596	0.0998	-5.6047	0.0000
High	AlcoholRiskMedium/High risk	0.9016	0.1182	7.6270	0.0000

High	SmokingDurationLess than 10 years	0.1171	0.1602	0.7310	0.4648
High	SmokingDuration10 to 19 years	0.1778	0.1513	1.1753	0.2399
High	SmokingDuration20 to 29 years	0.4380	0.1438	3.0458	0.0023
High	SmokingDuration30 to 39 years	0.4492	0.1573	2.8556	0.0043
High	SmokingDuration40 to 49 years	0.5857	0.1817	3.2229	0.0013
High	SmokingDuration50 years and over	0.4651	0.2920	1.5930	0.1112
High	NonSchoolQualsYes	-0.0662	0.0985	-0.6722	0.5014
High	IncomeDecile2	-0.4006	0.2121	-1.8885	0.0590
High	IncomeDecile3	-0.6140	0.2122	-2.8939	0.0038
High	IncomeDecile4	-0.3944	0.2065	-1.9098	0.0562
High	IncomeDecile5	-0.5560	0.2223	-2.5011	0.0124
High	IncomeDecile6	-0.3547	0.2193	-1.6175	0.1058
High	IncomeDecile7	-0.7346	0.2313	-3.1759	0.0015
High	IncomeDecile8	-0.3768	0.2223	-1.6948	0.0901
High	IncomeDecile9	-0.6175	0.2275	-2.7147	0.0066
High	IncomeDecile10	-0.3185	0.2256	-1.4116	0.1581
High	Age	0.0118	0.0034	3.4843	0.0005
High	Waist	0.0398	0.0032	12.4575	0.0000

It is significant for High GGT but still not significant for Borderline GGT in this case, so there is some slight association. We now try fitting some linear models as well.

Linear Model Fitting

▼ Code

```
# ALT Linear Model
ALT.linear = lm(ALT ~ ExerciseGuidelines + Sex + AlcoholRisk +
                          SmokingDuration + NonSchoolQuals + In
                          Age + Waist, data = dat.full)
tidy(ALT.linear) %>%
  kable(caption = "ALT Linear Regression Summary Table",
        digits = 4, align = "cccccc")
```

ALT Linear Regression Summary Table

term	estimate	std.error	statistic	p.value

(Intercept)	1.6023	1.1753	1.3633	0.1728
ExerciseGuidelinesNo	0.1805	0.3045	0.5927	0.5534
SexMale	5.4871	0.3267	16.7959	0.0000
AlcoholRiskMedium/High risk	1.5153	0.4508	3.3613	0.0008
SmokingDurationLess than 10 years	0.0891	0.4943	0.1803	0.8569
SmokingDuration10 to 19 years	0.6095	0.4797	1.2706	0.2039
SmokingDuration20 to 29 years	0.0972	0.5144	0.1889	0.8502
SmokingDuration30 to 39 years	0.2166	0.5931	0.3653	0.7149
SmokingDuration40 to 49 years	0.3268	0.7105	0.4600	0.6455
SmokingDuration50 years and over	-4.0632	1.1390	-3.5675	0.0004
NonSchoolQualsYes	0.8113	0.3301	2.4575	0.0140
IncomeDecile2	-1.6619	0.7575	-2.1938	0.0283
IncomeDecile3	-1.3517	0.7453	-1.8137	0.0698
IncomeDecile4	-0.7829	0.7312	-1.0708	0.2843
IncomeDecile5	-0.8588	0.7577	-1.1333	0.2571
IncomeDecile6	-0.7797	0.7581	-1.0285	0.3038
IncomeDecile7	-1.1005	0.7512	-1.4650	0.1430
IncomeDecile8	0.0871	0.7517	0.1159	0.9077
IncomeDecile9	0.1118	0.7377	0.1515	0.8796
IncomeDecile10	0.1399	0.7606	0.1839	0.8541
Age	-0.0737	0.0104	-7.0829	0.0000
Waist	0.2284	0.0109	20.8589	0.0000

▼ Code

```
# GGT Linear Model
GGT.linear = lm(GGT ~ ExerciseGuidelines + Sex + AlcoholRisk +
                        SmokingDuration + NonSchoolQuals + In
                        Age + Waist, data = dat.full)
tidy(GGT.linear) %>%
  kable(caption = "GGT Linear Regression Summary Table",
        digits = 4, align = "ccccc")
```

GGT Linear Regression Summary Table

term	estimate	std.error	statistic	p.value
(Intercept)	-7.7397	1.6026	-4.8296	0.0000

ExerciseGuidelinesNo	1.5997	0.4153	3.8524	0.0001
SexMale	4.8283	0.4455	10.8386	0.0000
AlcoholRiskMedium/High risk	6.1408	0.6147	9.9899	0.0000
SmokingDurationLess than 10 years	0.5902	0.6740	0.8756	0.3813
SmokingDuration10 to 19 years	1.6178	0.6541	2.4734	0.0134
SmokingDuration20 to 29 years	2.8788	0.7014	4.1043	0.0000
SmokingDuration30 to 39 years	3.3489	0.8087	4.1410	0.0000
SmokingDuration40 to 49 years	4.8288	0.9688	4.9844	0.0000
SmokingDuration50 years and over	4.7410	1.5531	3.0527	0.0023
NonSchoolQualsYes	-0.4080	0.4501	-0.9065	0.3647
IncomeDecile2	-1.8181	1.0330	-1.7601	0.0785
IncomeDecile3	-2.7816	1.0162	-2.7372	0.0062
IncomeDecile4	-2.0328	0.9970	-2.0390	0.0415
IncomeDecile5	-2.8592	1.0332	-2.7672	0.0057
IncomeDecile6	-1.3737	1.0337	-1.3289	0.1840
IncomeDecile7	-3.3585	1.0244	-3.2787	0.0011
IncomeDecile8	-1.9132	1.0250	-1.8666	0.0620
IncomeDecile9	-2.4914	1.0059	-2.4767	0.0133
IncomeDecile10	-1.1203	1.0371	-1.0802	0.2801
Age	0.0424	0.0142	2.9893	0.0028
Waist	0.3001	0.0149	20.1030	0.0000

We do see a similar trend here where the effect of exercise is not significant in ALT but it is significant in GGT.

Model Diagnostics

We next conduct model diagnostics and evaluation, especially for the case of linear regression models where it is easier to check assumptions. We check collinearity of the four models; the multinomial logistic regression models will reuse the helper function I made back in `ModelDiagnosticsQ2`, and for linear regression models we stick to regular VIF.

▼ Code

```
# Helper function
rk.collinear.check = function(model, model_name = NULL) {
```

```
  X = model.matrix(model)
  X_no_intercept = X[, colnames(X) != "(Intercept)", drop = FALS
  qr_rank = qr(X)$rank
  n_cols = ncol(X)

  data.frame(model = ifelse(is.null(model_name), "", model_name)
             n_obs = nrow(X),
             n_columns = n_cols,
             qr_rank = qr_rank,
             rank_deficiency = n_cols - qr_rank,
             full_rank = qr_rank == n_cols,
             kappa = kappa(X_no_intercept))

}
r1 = rk.collinear.check(ALT.multinom, "ALT3")
r2 = rk.collinear.check(GGT.multinom, "GGT3")
rbind(r1, r2) %>% kable(
  caption = "Multinomial Logistic Regression Models Collinearity
  align = "ccccccc")
```

Multinomial Logistic Regression Models Collinearity Check

model	n_obs	n_columns	qr_rank	rank_deficiency	full_rank	kappa
ALT3	4762	22	22	0	TRUE	3831.121
GGT3	4762	22	22	0	TRUE	3831.121

▼ Code

```
# VIF of linear models
vif(ALT.linear) %>% kable(
  caption = "ALT Linear Model VIF", digits = 4, align = "ccc"
)
```

ALT Linear Model VIF

	GVIF	Df	GVIF^(1/(2*Df))
ExerciseGuidelines	1.0729	1	1.0358
Sex	1.2321	1	1.1100
AlcoholRisk	1.0472	1	1.0233
SmokingDuration	1.2264	6	1.0172
NonSchoolQuals	1.1532	1	1.0739

IncomeDecile	1.4502	9	1.0209
Age	1.3379	1	1.1567
Waist	1.2542	1	1.1199

▼ Code

```
vif(GGT.linear) %>% kable(
  caption = "GGT Linear Model VIF", digits = 4, align = "ccc"
)
```

GGT Linear Model VIF

	GVIF	Df	GVIF^(1/(2*Df))
ExerciseGuidelines	1.0729	1	1.0358
Sex	1.2321	1	1.1100
AlcoholRisk	1.0472	1	1.0233
SmokingDuration	1.2264	6	1.0172
NonSchoolQuals	1.1532	1	1.0739
IncomeDecile	1.4502	9	1.0209
Age	1.3379	1	1.1567
Waist	1.2542	1	1.1199

Overall, there is no apparent multicollinearity this time around, likely since we turned some variables to numeric such as age, in this case the models are in general seemingly more stable overall. Next, we run cross-tabulations between our different covariates and `ALT3` / `GGT3` to ensure there are no sparse categories.

▼ Code

```
# Sparseness check for ALT3
xtabs(~ ALT3 + AlcoholRisk, data = model.frame(ALT.multinom))
```

```
            AlcoholRisk
ALT3         Low risk Medium/High risk
  Low             3219               439
  Borderline       452                73
  High             484                95
```

▼ Code

```
xtabs(~ ALT3 + SmokingDuration, data = model.frame(ALT.multinom
```

```
            SmokingDuration
ALT3         Never smoked daily Less than 10 years 10 to 19
years
  Low                      2007                379
404
  Borderline                263                 73
68
  High                      268                 60
84
            SmokingDuration
ALT3         20 to 29 years 30 to 39 years 40 to 49 years 50
years and over
  Low                   354            262            177
75
  Borderline             55             35             25
6
  High                   70             55             36
6
```

▼ Code

```
xtabs(~ ALT3 + ExerciseGuidelines, data = model.frame(ALT.multi
```

```
            ExerciseGuidelines
ALT3          Yes   No
  Low        1913 1745
  Borderline  286  239
  High        265  314
```

▼ Code

```
xtabs(~ ALT3 + Sex, data = model.frame(ALT.multinom))
```

```
            Sex
ALT3         Female Male
  Low          2097 1561
  Borderline    181  344
  High          241  338
```

▼ Code

```
xtabs(~ ALT3 + NonSchoolQuals, data = model.frame(ALT.multinom)
```

```
            NonSchoolQuals
ALT3           No  Yes
```

```
  Low        1325 2333
  Borderline  163  362
  High        193  386
```

▼ Code

```
xtabs(~ ALT3 + IncomeDecile, data = model.frame(ALT.multinom))
```

```
           IncomeDecile
ALT3          1   2   3   4   5   6   7   8   9  10
  Low       232 376 448 447 345 342 368 351 397 352
  Borderline  28  45  27  60  41  54  50  63  79  78
  High        41  52  64  53  62  52  60  68  70  57
```

▼ Code

```
# Sparseness check for GGT3
xtabs(~ GGT3 + AlcoholRisk, data = model.frame(GGT.multinom))
```

```
           AlcoholRisk
GGT3         Low risk Medium/High risk
  Low            3453              417
  Borderline      215               53
  High            487              137
```

▼ Code

```
xtabs(~ GGT3 + SmokingDuration, data = model.frame(GGT.multinom
```

```
           SmokingDuration
GGT3         Never smoked daily Less than 10 years 10 to 19
years
  Low                      2163                417
443
  Borderline                112                 36
44
  High                      263                 59
69
           SmokingDuration
GGT3         20 to 29 years 30 to 39 years 40 to 49 years 50
years and over
  Low                   360            261            169
57
  Borderline             31             19             15
11
  High                   88             72             54
```

19

▼ Code

```
xtabs(~ GGT3 + ExerciseGuidelines, data = model.frame(GGT.multi
```

```
           ExerciseGuidelines
GGT3          Yes   No
  Low        2083 1787
  Borderline  128  140
  High        253  371
```

▼ Code

```
xtabs(~ GGT3 + Sex, data = model.frame(GGT.multinom))
```

```
           Sex
GGT3        Female Male
  Low         2049 1821
  Borderline   124  144
  High         346  278
```

▼ Code

```
xtabs(~ GGT3 + NonSchoolQuals, data = model.frame(GGT.multinom)
```

```
           NonSchoolQuals
GGT3          No  Yes
  Low        1315 2555
  Borderline  101  167
  High        265  359
```

▼ Code

```
xtabs(~ GGT3 + IncomeDecile, data = model.frame(GGT.multinom))
```

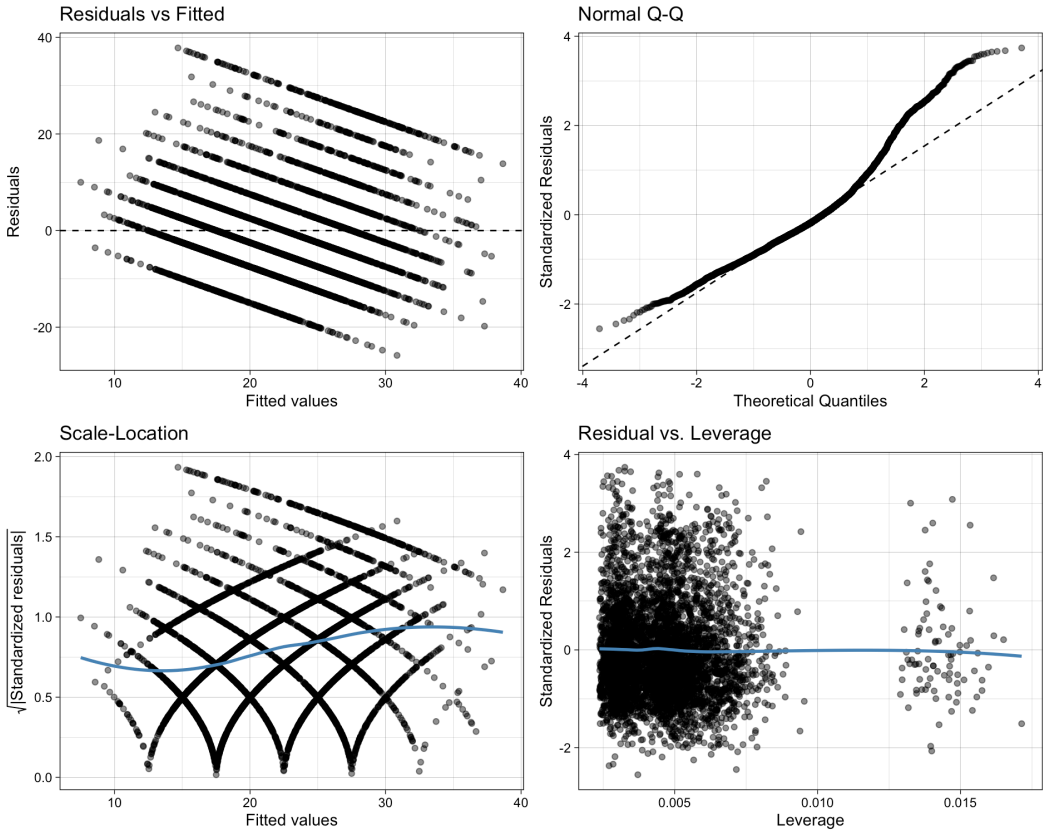
```
           IncomeDecile
GGT3          1   2   3   4   5   6   7   8   9  10
  Low       237 363 423 447 366 373 401 391 464 405
  Borderline  14  28  37  26  27  15  31  33  31  26
  High        50  82  79  87  55  60  46  58  51  56
```

Smoking duration may be a bit sparse for `50 years and over`, so I may go back and collapse that level at some point. For now, we check the residuals of the

linear model for the heteroskedasticity assumption, uncorrelatedness assumption, linearity assumption, and normality assumption; we also can check high-leverage and influential points as well.

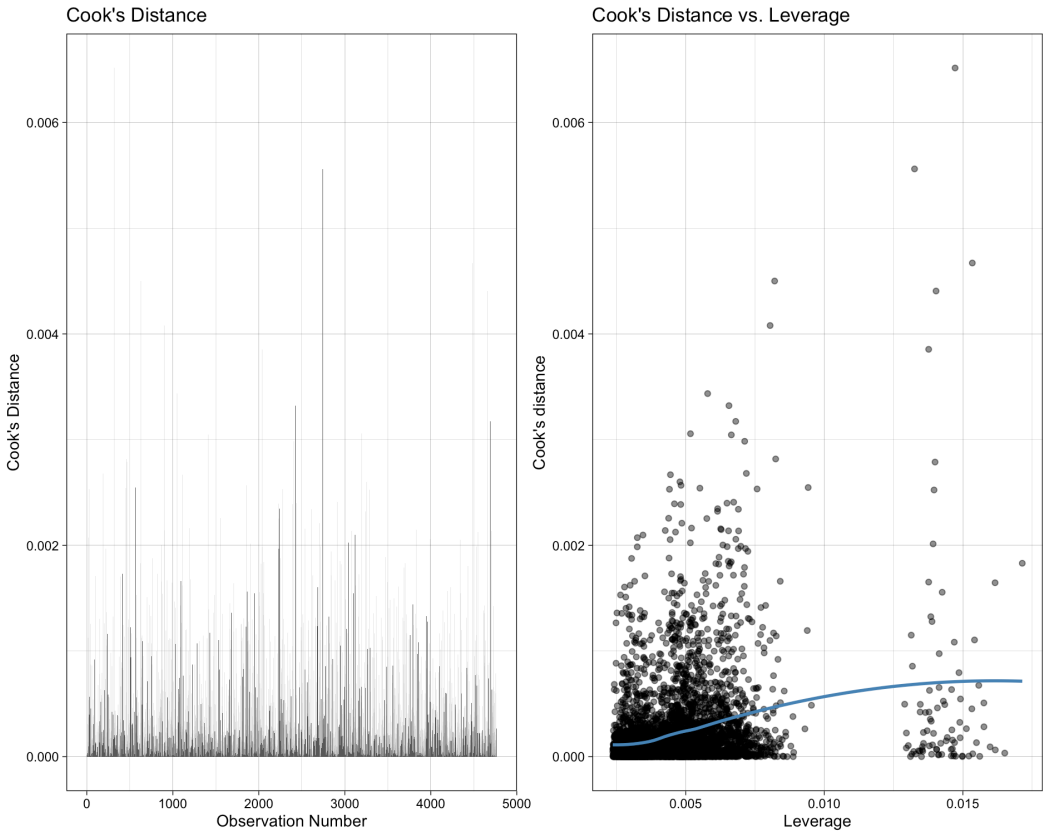
▼ Code

```
# ALT residual diagnostics
gglm(ALT.linear) & theme_linedraw()
```



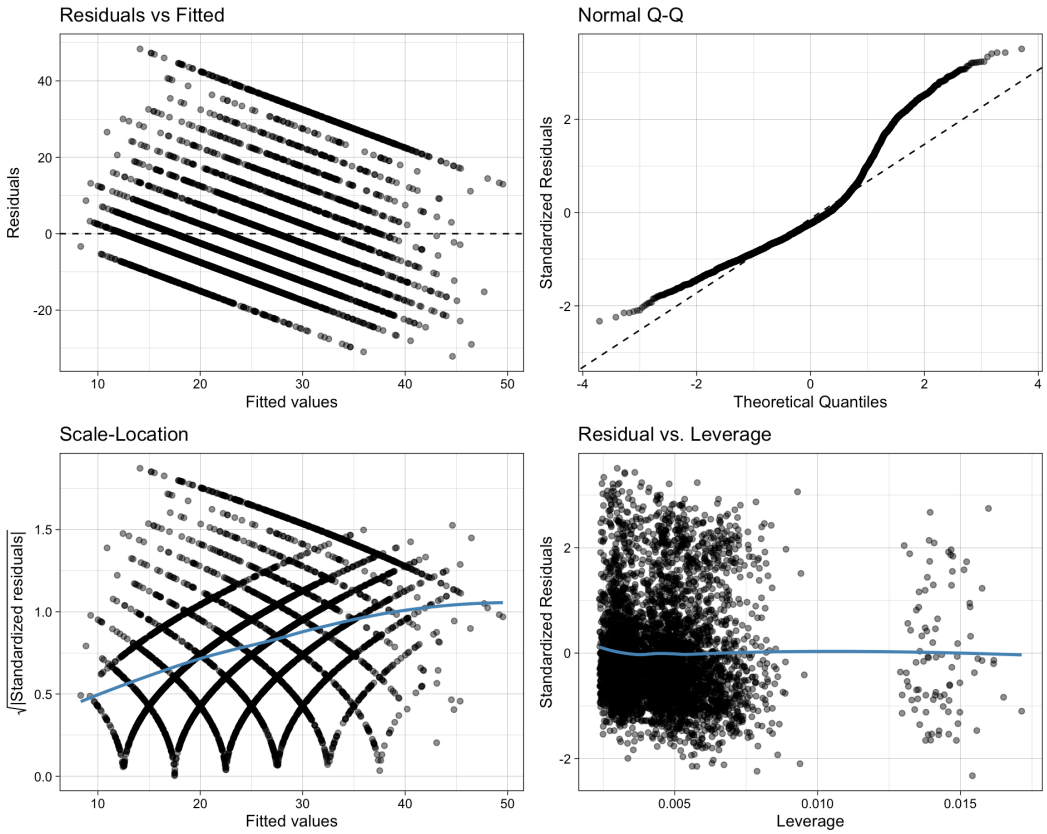
▼ Code

```
p1 = ggplot(ALT.linear) + stat_cooks_obs() + theme_linedraw()
p2 = ggplot(ALT.linear) + stat_cooks_leverage() + theme_linedrav
p1 + p2
```



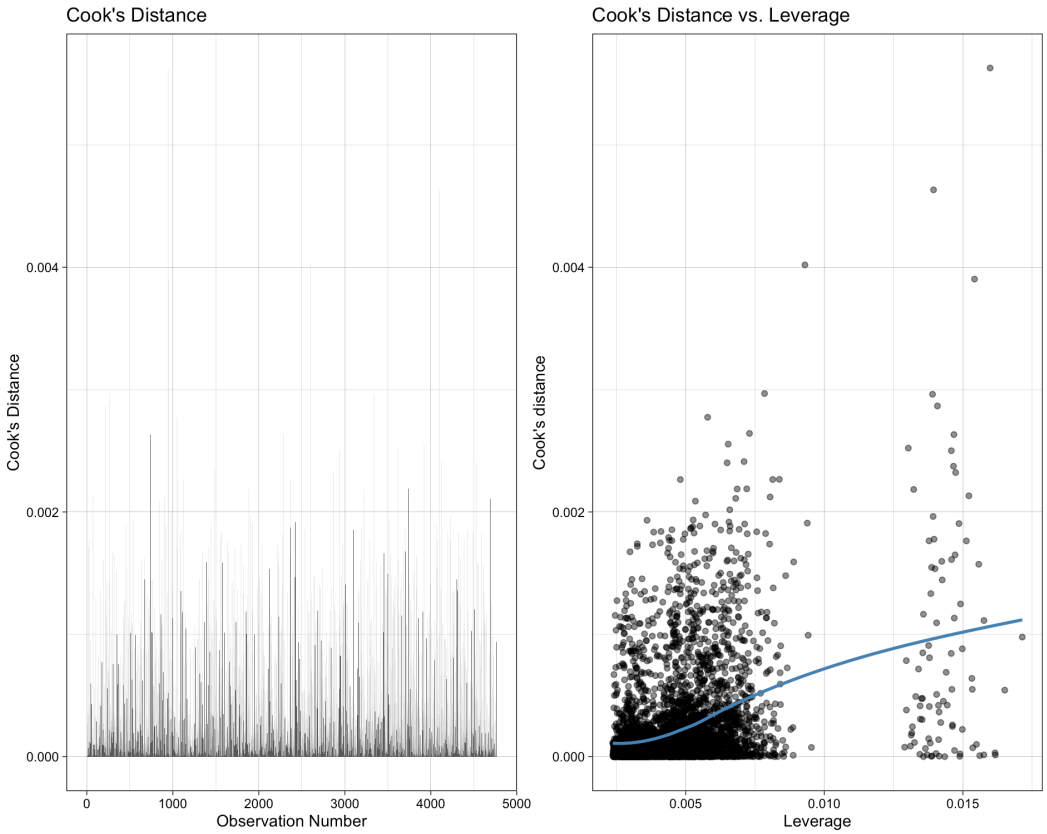
▼ Code

```
# GGT residual diagnostics
gglm(GGT.linear) & theme_linedraw()
```



▼ Code

```
p1 = ggplot(GGT.linear) + stat_cooks_obs() + theme_linedraw()
p2 = ggplot(GGT.linear) + stat_cooks_leverage() + theme_linedrav
p1 + p2
```



Unfortunately, it seems that the linear models are not as well-behaved as we hope, since several fundamental assumptions are clearly violated here. We can take the logarithms of ALT and GGT to see if that helps.

▼ Code

```
# Log(ALT/GGT) models
ALT.loglinear = lm(log(ALT) ~ExerciseGuidelines + Sex + AlcoholF
                     SmokingDuration + NonSchoolQuals + IncomeDe
                   data = dat.full)
GGT.loglinear = lm(log(GGT) ~ExerciseGuidelines + Sex + AlcoholF
                     SmokingDuration + NonSchoolQuals + IncomeDe
                   data = dat.full)
tidy(ALT.loglinear) %>%
  kable(caption = "Log ALT Linear Regression Summary Table",
        digits = 4, align = "cccccc")
```

Log ALT Linear Regression Summary Table

term	estimate	std.error	statistic	p.value
(Intercept)	1.9937	0.0553	36.0636	0.0000
ExerciseGuidelinesNo	-0.0001	0.0143	-0.0081	0.9935

SexMale	0.2649	0.0154	17.2368	0.0000
AlcoholRiskMedium/High risk	0.0674	0.0212	3.1799	0.0015
SmokingDurationLess than 10 years	-0.0042	0.0233	-0.1810	0.8564
SmokingDuration10 to 19 years	0.0137	0.0226	0.6093	0.5423
SmokingDuration20 to 29 years	-0.0181	0.0242	-0.7460	0.4557
SmokingDuration30 to 39 years	-0.0214	0.0279	-0.7662	0.4436
SmokingDuration40 to 49 years	-0.0174	0.0334	-0.5206	0.6027
SmokingDuration50 years and over	-0.2571	0.0536	-4.7993	0.0000
NonSchoolQualsYes	0.0381	0.0155	2.4518	0.0142
IncomeDecile2	-0.0637	0.0356	-1.7887	0.0737
IncomeDecile3	-0.0689	0.0351	-1.9662	0.0493
IncomeDecile4	-0.0185	0.0344	-0.5389	0.5900
IncomeDecile5	-0.0288	0.0356	-0.8074	0.4195
IncomeDecile6	-0.0144	0.0357	-0.4027	0.6872
IncomeDecile7	-0.0252	0.0353	-0.7124	0.4762
IncomeDecile8	0.0041	0.0354	0.1172	0.9067
IncomeDecile9	0.0323	0.0347	0.9308	0.3520
IncomeDecile10	0.0455	0.0358	1.2712	0.2037
Age	-0.0020	0.0005	-3.9836	0.0001
Waist	0.0099	0.0005	19.2753	0.0000

▼ Code

```
tidy(GGT.loglinear) %>%
  kable(caption = "Log GGT Linear Regression Summary Table",
        digits = 4, align = "cccccc")
```

Log GGT Linear Regression Summary Table

term	estimate	std.error	statistic	p.value
(Intercept)	1.6164	0.0623	25.9404	0.0000
ExerciseGuidelinesNo	0.0610	0.0161	3.7765	0.0002
SexMale	0.2337	0.0173	13.4903	0.0000
AlcoholRiskMedium/High risk	0.2180	0.0239	9.1218	0.0000
SmokingDurationLess than 10 years	0.0213	0.0262	0.8121	0.4168
SmokingDuration10 to 19 years	0.0707	0.0254	2.7813	0.0054

SmokingDuration20 to 29 years	0.1136	0.0273	4.1659	0.0000
SmokingDuration30 to 39 years	0.1204	0.0314	3.8287	0.0001
SmokingDuration40 to 49 years	0.1727	0.0377	4.5835	0.0000
SmokingDuration50 years and over	0.1560	0.0604	2.5836	0.0098
NonSchoolQualsYes	-0.0155	0.0175	-0.8882	0.3745
IncomeDecile2	-0.0494	0.0402	-1.2292	0.2191
IncomeDecile3	-0.0671	0.0395	-1.6987	0.0894
IncomeDecile4	-0.0461	0.0388	-1.1897	0.2342
IncomeDecile5	-0.0735	0.0402	-1.8296	0.0674
IncomeDecile6	-0.0094	0.0402	-0.2328	0.8159
IncomeDecile7	-0.0832	0.0398	-2.0888	0.0368
IncomeDecile8	-0.0393	0.0399	-0.9869	0.3237
IncomeDecile9	-0.0643	0.0391	-1.6440	0.1002
IncomeDecile10	-0.0045	0.0403	-0.1119	0.9109
Age	0.0024	0.0006	4.3796	0.0000
Waist	0.0123	0.0006	21.1823	0.0000

▼ Code

```
# VIF
vif(ALT.loglinear) %>% kable(
  caption = "Log ALT Linear Model VIF", digits = 4, align = "cc
)
```

Log ALT Linear Model VIF

	GVIF	Df	GVIF^(1/(2*Df))
ExerciseGuidelines	1.0729	1	1.0358
Sex	1.2321	1	1.1100
AlcoholRisk	1.0472	1	1.0233
SmokingDuration	1.2264	6	1.0172
NonSchoolQuals	1.1532	1	1.0739
IncomeDecile	1.4502	9	1.0209
Age	1.3379	1	1.1567
Waist	1.2542	1	1.1199

▼ Code

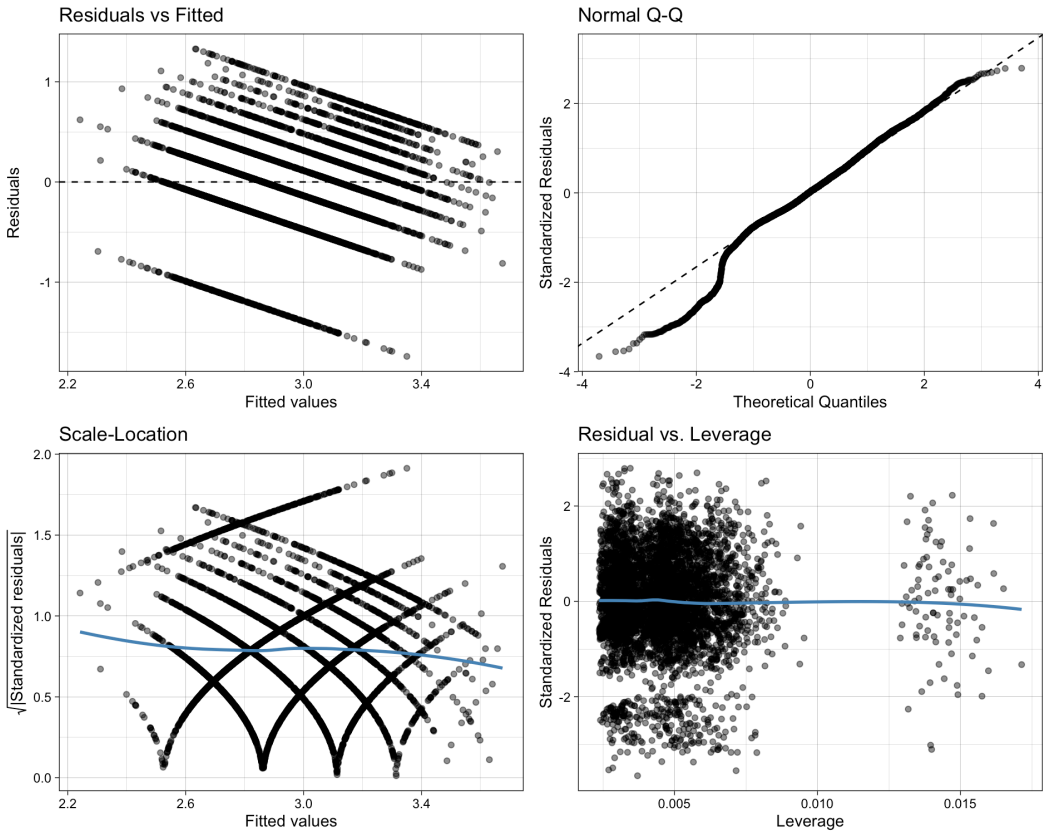
```
vif(GGT.loglinear) %>% kable(
  caption = "Log GGT Linear Model VIF", digits = 4, align = "cc
)
```

Log GGT Linear Model VIF

	GVIF	Df	GVIF^(1/(2*Df))
ExerciseGuidelines	1.0729	1	1.0358
Sex	1.2321	1	1.1100
AlcoholRisk	1.0472	1	1.0233
SmokingDuration	1.2264	6	1.0172
NonSchoolQuals	1.1532	1	1.0739
IncomeDecile	1.4502	9	1.0209
Age	1.3379	1	1.1567
Waist	1.2542	1	1.1199

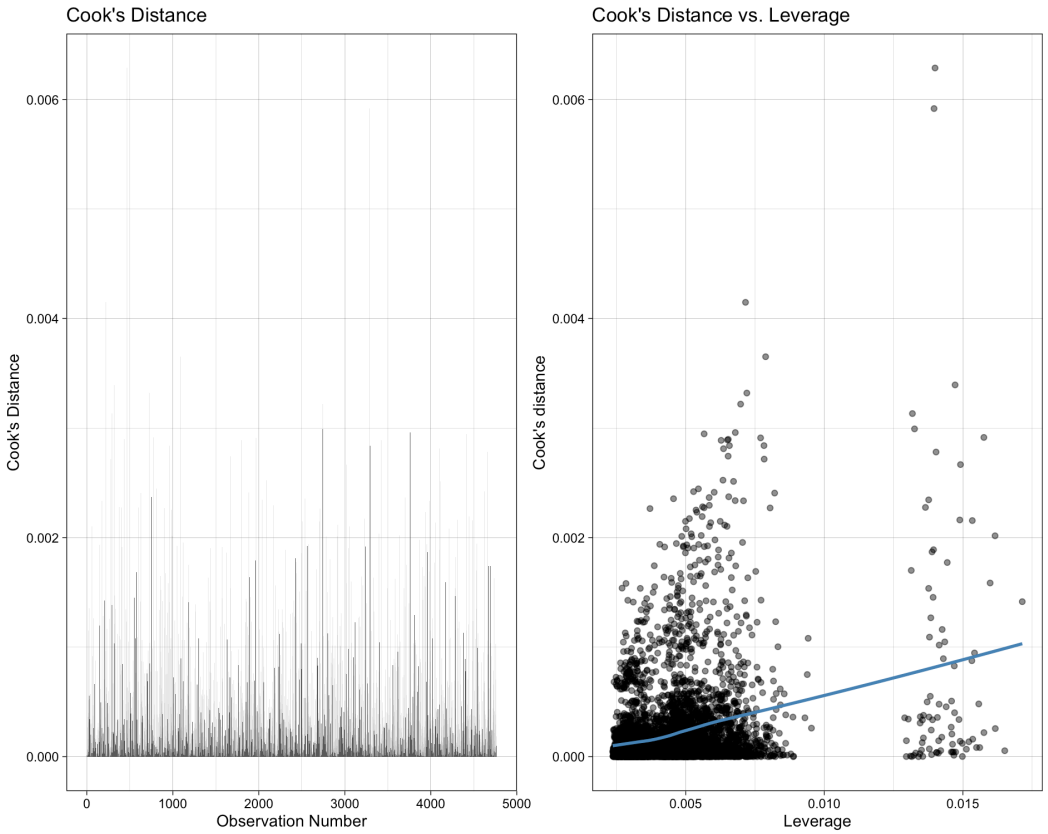
▼ Code

```
# Log ALT residual diagnostics
gglm(ALT.loglinear) & theme_linedraw()
```



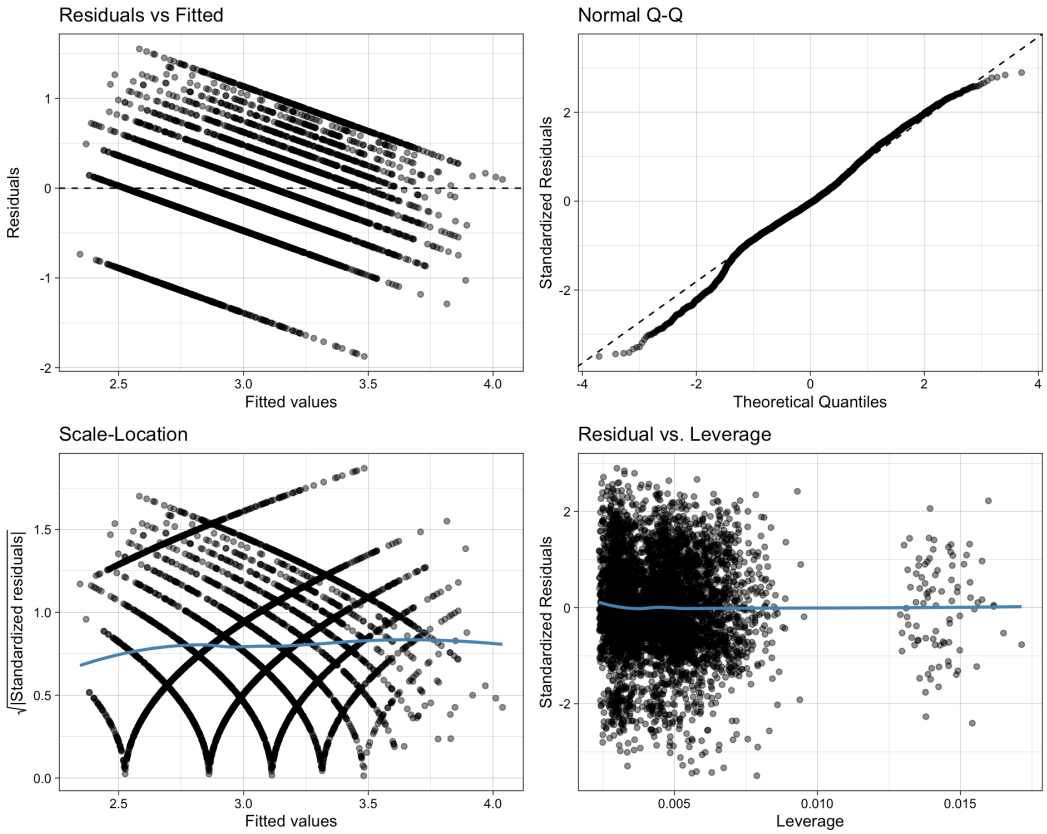
▼ Code

```
p1 = ggplot(ALT.loglinear) + stat_cooks_obs() + theme_linedraw()
p2 = ggplot(ALT.loglinear) + stat_cooks_leverage() + theme_linedr
p1 + p2
```



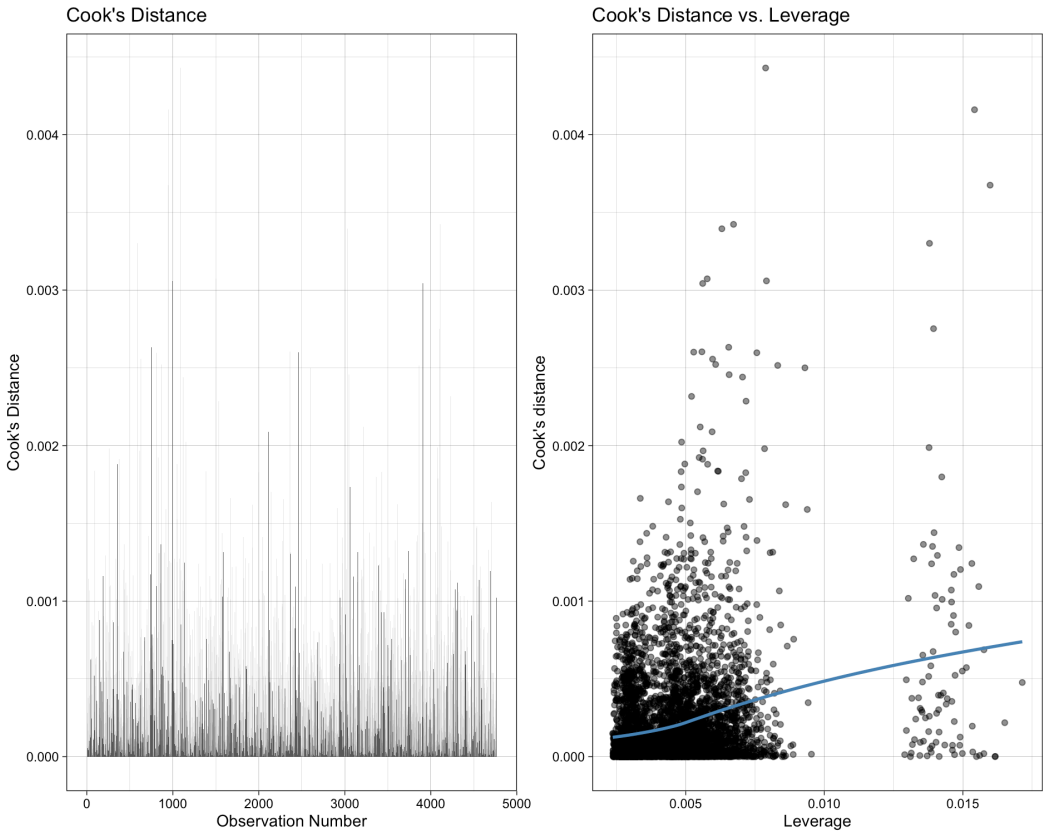
▼ Code

```
# Log GGT residual diagnostics
gglm(GGT.loglinear) & theme_linedraw()
```



▼ Code

```
p1 = ggplot(GGT.loglinear) + stat_cooks_obs() + theme_linedraw(
p2 = ggplot(GGT.loglinear) + stat_cooks_leverage() + theme_linec
p1 + p2
```



This tells us the log transformation does improve on the heteroskedasticity and most of the other issues, but it does reveal more clearly the two distinct groups that `ALT` and `GGT` are distributed as; this makes sense since we originally had Low ALT/GGT as the majority and Borderline/High as a small minority.

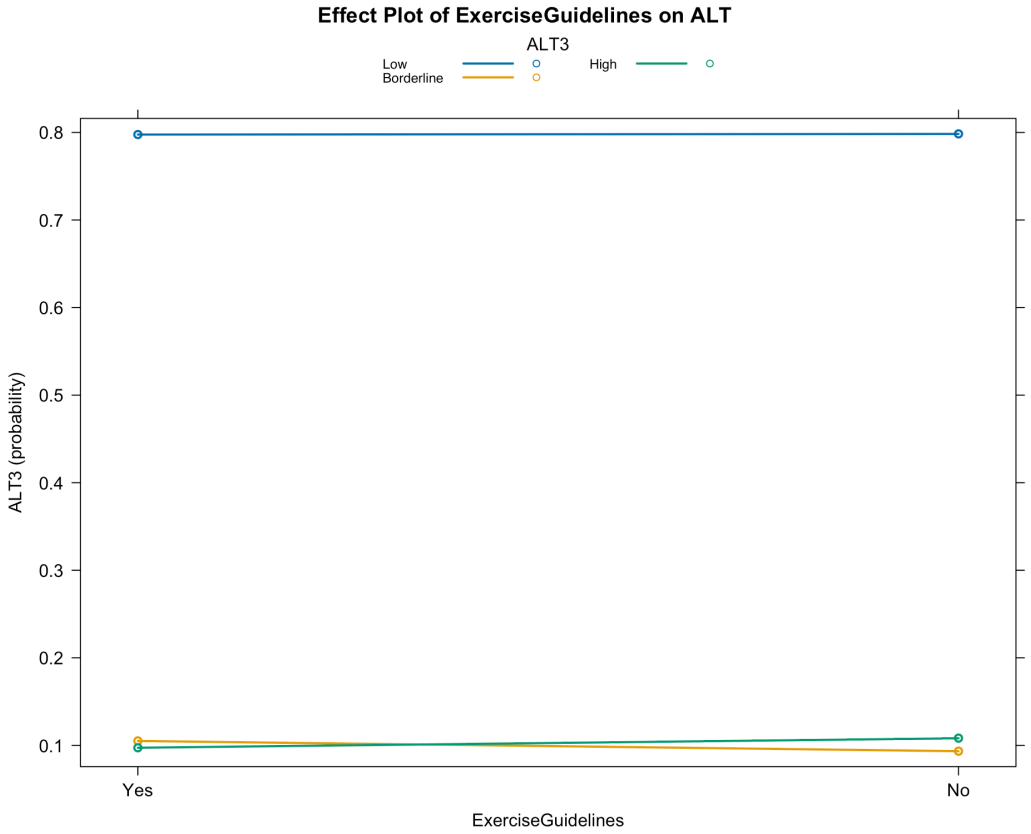
Overall, I think we can used the log-transformed linear models as a supplement, as while the results are largely the same, using the continuous form of ALT and GGT is likely more powerful for our case.

Marginal Effects of Multinomial Logistic Models

We have done this in `CausalInference`, so I thought we can do the same as well here.

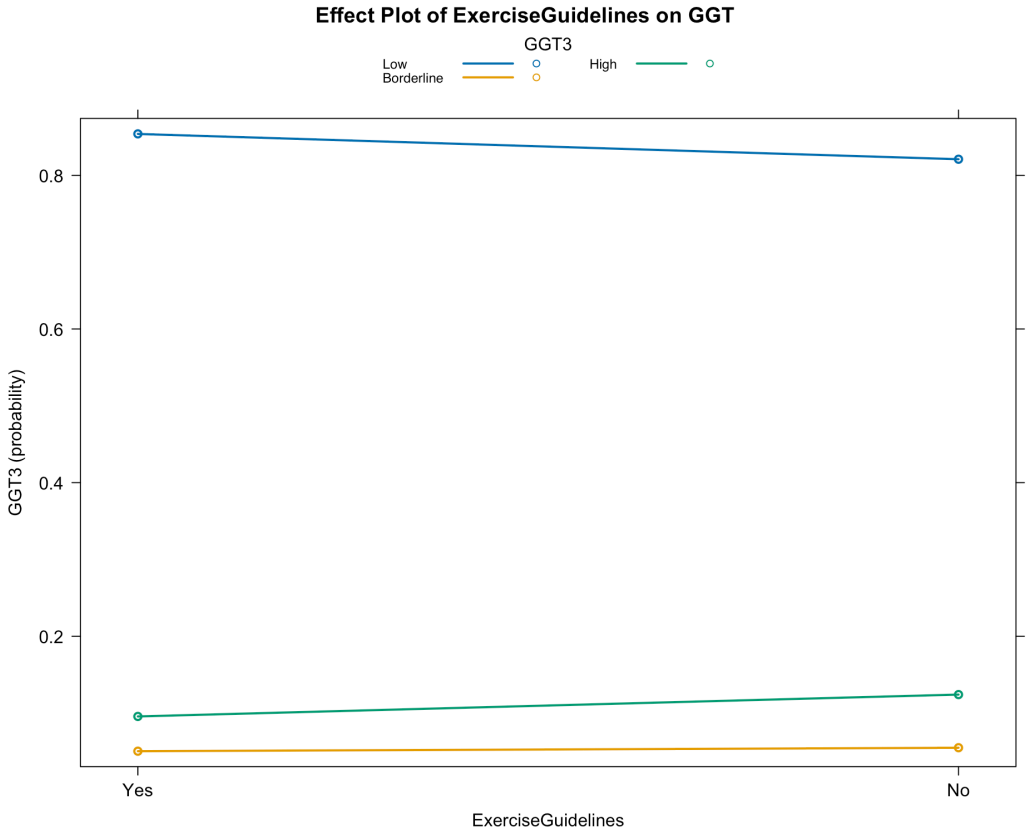
▼ Code

```
# Effects
eff.alt = effect("ExerciseGuidelines", ALT.multinom)
plot(eff.alt, multiline = TRUE, main = "Effect Plot of Exercise(
```



▼ Code

```
eff.ggt = effect("ExerciseGuidelines", GGT.multinom)
plot(eff.ggt, multiline = TRUE, main = "Effect Plot of Exercise(
```



MICE Imputation Sensitivity Analysis

Again, we have done this in `SensitivityAnalyses`, so I want to try again here with the updated data set.

▼ Code

```
# dat with all observations
dat.all = dat %>% mutate(
  AL2K7DAY = case_when(AL2K7DAY %in% c("Medium risk", "High risk
                       ~ "Medium/High risk",
                       .default = "Low risk") %>% as.factor(),
  SMKAGEDB = fct_relevel(SMKAGEDB, "Never smoked daily", "Less t
                         "10 to 19 years", "20 to 29 years", "30
                         "40 to 49 years", "50 years and over"),
  EXREGUIC = fct_recode(EXREGUIC,
                        "Yes" = "Met recommended guidelines",
                        "No" = "Did not meet recommended guideli
  EDATTQ2 = fct_recode(EDATTQ2,
                       "No" = "Does not have non-school qualific
                       "Yes" = "Has non-school qualification"),
  ) %>% rename(
    "AlcoholRisk" = AL2K7DAY,
```

```
    "SmokingDuration" = SMKAGEDB,
    "ExerciseGuidelines" = EXREGUIC,
    "NonSchoolQuals" = EDATTQ2,
    "Sex" = SEX,
    "Age" = AGE,
    "IncomeDecile" = INCDECPN,
    "Waist" = WAIST,
    "ALT10" = ALTRESB,
    "GGT12" = GGTRESB
  ) %>% dplyr::select(-c(ABSLID, ABSPID))

file.name = here("Objects/dat_imputed.rds")
if (file.exists(file.name)) {
  dat.imp = readRDS(file.name)
} else {
  dat.imp = mice(dat.all, m = 10, maxit = 10, method = "pmm", pr
  saveRDS(dat.imp, file.name)
}
```

Next, we pool the results for ALT and then for GGT.

▼ Code

```
# Pooled ALT
ALT.multinom.mice = with(dat.imp,
                     expr = nnet::multinom(formula = ALT3 ~ Exer
                                           Sex +
                                       AlcoholRisk +
                                       SmokingDuration + NonSchc
                                       IncomeDecile + Age + Wais
ALT.multinom.mice.pooled = pool(ALT.multinom.mice)
df = data.frame(summary(ALT.multinom.mice.pooled))
Alt.Exr1 = df[c(2, 24), ] %>%
  mutate("y.level" = c("Borderline", "High")) %>%
  relocate(last_col()) %>% dplyr::select(-c(term, df))
Alt.Exr2 = tidy(ALT.multinom)[c(2, 24), ] %>%
  dplyr::select(-c(term))
ALT.rows = rbind(Alt.Exr1, Alt.Exr2) %>%
  mutate("Model" = rep(c("MICE", "Original"), each = 2)) %>%
  relocate(last_col())
rownames(ALT.rows) = NULL
kable(ALT.rows, caption = "Sensitivity Analysis for MICE on ALT'
      digits = 4, align = "cccccc",
      col.names = c("Model", "ALT Level", "Coefficient",
                    "SE", "t-Score", "p-value"))
```

Sensitivity Analysis for MICE on ALT

Model	ALT Level	Coefficient	SE	t-Score	p-value
MICE	Borderline	-0.0348	0.0794	-0.4382	0.6659
MICE	High	0.0565	0.0816	0.6921	0.4961
Original	Borderline	-0.1195	0.0993	-1.2031	0.2289
Original	High	0.1059	0.0966	1.0969	0.2727

▼ Code

```
# Pooled GGT
GGT.multinom.mice = with(dat.imp,
                     expr = nnet::multinom(formula = GGT3 ~ Exe
                                           Sex +
                                       AlcoholRisk +
                                       SmokingDuration + NonSch
                                       IncomeDecile + Age + Wai
GGT.multinom.mice.pooled = pool(GGT.multinom.mice)
df = data.frame(summary(GGT.multinom.mice.pooled))
GGT.Exr1 = df[c(2, 24), ] %>%
  mutate("y.level" = c("Borderline", "High")) %>%
  relocate(last_col()) %>% dplyr::select(-c(term, df))
GGT.Exr2 = tidy(GGT.multinom)[c(2, 24), ] %>%
  dplyr::select(-c(term))
GGT.rows = rbind(GGT.Exr1, GGT.Exr2) %>%
  mutate("Model" = rep(c("MICE", "Original"), each = 2)) %>%
  relocate(last_col())
rownames(GGT.rows) = NULL
kable(GGT.rows, caption = "Sensitivity Analysis for MICE on GGT"
      digits = 4, align = "cccccc",
      col.names = c("Model", "GGT Level", "Coefficient",
                    "SE", "t-Score", "p-value"))
```

Sensitivity Analysis for MICE on GGT

Model	GGT Level	Coefficient	SE	t-Score	p-value
MICE	Borderline	0.1348	0.0864	1.5605	0.1270
MICE	High	0.1879	0.0683	2.7499	0.0105
Original	Borderline	0.1258	0.1320	0.9524	0.3409
Original	High	0.2999	0.0939	3.1959	0.0014

Overall MICE does not change significance for either ALT and GGT (i.e., significant results stay significant and likewise for non-significant results). It does suggest our

model is robust to missing data, which is reassuring in that sense.

Binary Logistic Regression

We can also check by combining Borderline and High for ALT and GGT as another sensitivity analysis, and in this case we can use the regular `glm` function.

▼ Code

```
# Add ALT2 and GGT2
dat.full = dat.full %>% mutate(
  ALT2 = case_when(ALT3 %in% c("Borderline", "High") ~ "High",
                   ALT3 == "Low" ~ "Low") %>% as.factor() %>% fc
  GGT2 = case_when(GGT3 %in% c("Borderline", "High") ~ "High",
                   GGT3 == "Low" ~ "Low") %>% as.factor() %>% fc
)
table(dat.full$ALT2); table(dat.full$GGT2)
```

```
 Low High
3658 1104
```

```
 Low High
3870  892
```

▼ Code

```
# Logistic regression
ALT.logistic = glm(ALT2 ~ ExerciseGuidelines + Sex + AlcoholRisk
    SmokingDuration + NonSchoolQuals + IncomeDecile + Age + Wais
    data = dat.full, family = "binomial")
GGT.logistic = glm(GGT2 ~ ExerciseGuidelines + Sex + AlcoholRisk
    SmokingDuration + NonSchoolQuals + IncomeDecile + Age + Wais
    data = dat.full, family = "binomial")

# Model summaries
tidy(ALT.logistic) %>% kable(caption = "Logistic Regression on A
                             align = "ccccc", digits = 4)
```

Logistic Regression on ALT2

term	estimate	std.error	statistic	p.value

(Intercept)	-4.8782	0.3021	-16.1463	0.0000
ExerciseGuidelinesNo	-0.0041	0.0748	-0.0546	0.9565
SexMale	0.4008	0.0786	5.1008	0.0000
AlcoholRiskMedium/High risk	0.2582	0.1058	2.4407	0.0147
SmokingDurationLess than 10 years	0.1952	0.1182	1.6519	0.0985
SmokingDuration10 to 19 years	0.2156	0.1143	1.8872	0.0591
SmokingDuration20 to 29 years	0.1119	0.1235	0.9062	0.3648
SmokingDuration30 to 39 years	0.1899	0.1417	1.3408	0.1800
SmokingDuration40 to 49 years	0.2621	0.1693	1.5482	0.1216
SmokingDuration50 years and over	-0.5549	0.3339	-1.6616	0.0966
NonSchoolQualsYes	0.1487	0.0817	1.8203	0.0687
IncomeDecile2	-0.3791	0.1906	-1.9885	0.0468
IncomeDecile3	-0.4241	0.1908	-2.2229	0.0262
IncomeDecile4	-0.1455	0.1829	-0.7957	0.4262
IncomeDecile5	-0.1331	0.1870	-0.7119	0.4765
IncomeDecile6	-0.0802	0.1858	-0.4314	0.6662
IncomeDecile7	-0.2958	0.1853	-1.5959	0.1105
IncomeDecile8	-0.0444	0.1824	-0.2435	0.8076
IncomeDecile9	-0.0507	0.1787	-0.2838	0.7766
IncomeDecile10	-0.1436	0.1827	-0.7860	0.4319
Age	-0.0151	0.0026	-5.7267	0.0000
Waist	0.0435	0.0027	16.0550	0.0000

▼ Code

```
tidy(GGT.logistic) %>% kable(caption = "Logistic Regression on (
                             align = "ccccc", digits = 4)
```

Logistic Regression on GGT2

term	estimate	std.error	statistic	p.value
(Intercept)	-5.2576	0.3274	-16.0599	0.0000
ExerciseGuidelinesNo	0.2462	0.0804	3.0635	0.0022
SexMale	-0.4079	0.0856	-4.7636	0.0000
AlcoholRiskMedium/High risk	0.8273	0.1036	7.9843	0.0000

SmokingDurationLess than 10 years	0.2291	0.1322	1.7322	0.0832
SmokingDuration10 to 19 years	0.2921	0.1249	2.3391	0.0193
SmokingDuration20 to 29 years	0.3934	0.1264	3.1117	0.0019
SmokingDuration30 to 39 years	0.3570	0.1418	2.5167	0.0118
SmokingDuration40 to 49 years	0.5056	0.1639	3.0844	0.0020
SmokingDuration50 years and over	0.6394	0.2503	2.5547	0.0106
NonSchoolQualsYes	-0.0668	0.0852	-0.7837	0.4332
IncomeDecile2	-0.2941	0.1900	-1.5479	0.1216
IncomeDecile3	-0.4019	0.1876	-2.1419	0.0322
IncomeDecile4	-0.3281	0.1858	-1.7660	0.0774
IncomeDecile5	-0.3761	0.1949	-1.9297	0.0536
IncomeDecile6	-0.3876	0.1982	-1.9554	0.0505
IncomeDecile7	-0.4768	0.1978	-2.4112	0.0159
IncomeDecile8	-0.2043	0.1933	-1.0566	0.2907
IncomeDecile9	-0.4305	0.1960	-2.1961	0.0281
IncomeDecile10	-0.2476	0.1981	-1.2503	0.2112
Age	0.0081	0.0029	2.8469	0.0044
Waist	0.0359	0.0028	12.9034	0.0000

Again, for ALT the exercise variable is highly non-significant but for GGT it is significant! We now check diagnostics as well as model comparison with our multinomial logistic regression.

▼ Code

```
# VIF
vif(ALT.logistic) %>% kable(
  caption = "ALT Logistic Regression Model VIF", digits = 4, al
)
```

ALT Logistic Regression Model VIF

	GVIF	Df	GVIF^(1/(2*Df))
ExerciseGuidelines	1.0715	1	1.0351
Sex	1.1562	1	1.0752
AlcoholRisk	1.0517	1	1.0255
SmokingDuration	1.2344	6	1.0177

NonSchoolQuals	1.1353	1	1.0655
IncomeDecile	1.4303	9	1.0201
Age	1.3352	1	1.1555
Waist	1.1956	1	1.0935

▼ Code

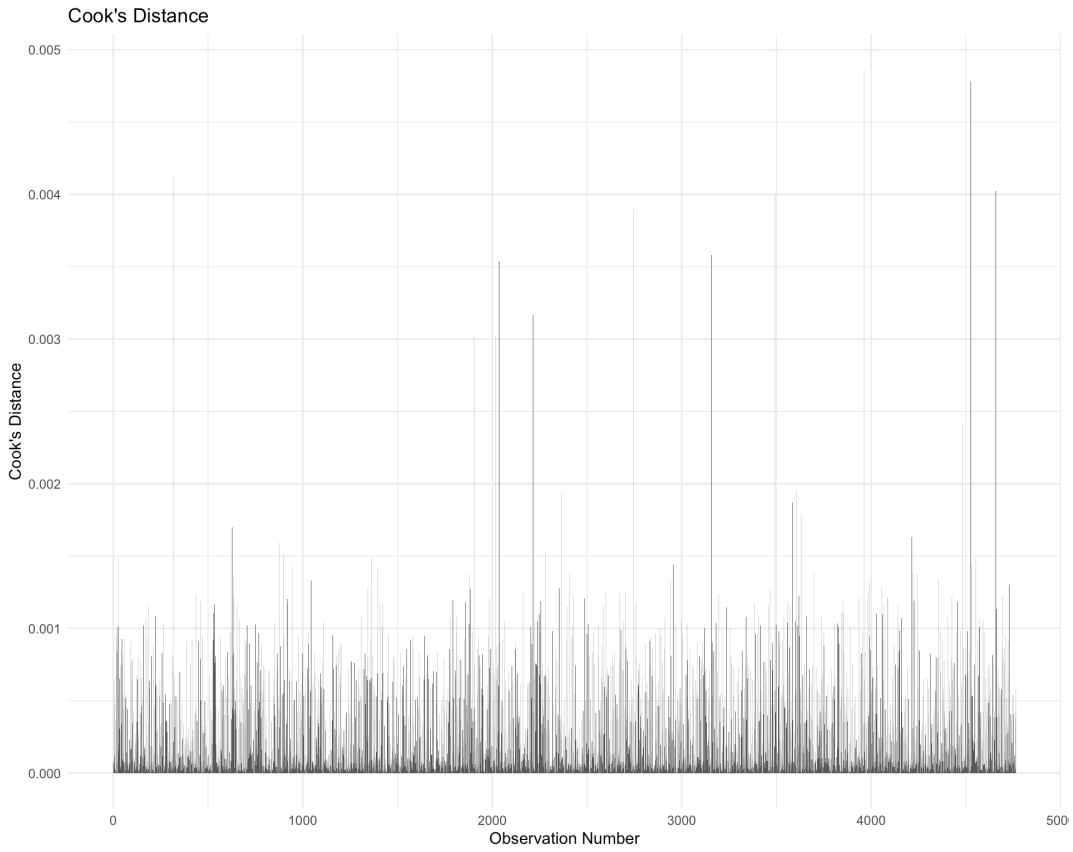
```
vif(GGT.logistic) %>% kable(
  caption = "GGT Logistic Regression Model VIF", digits = 4, ali
)
```

GGT Logistic Regression Model VIF

	GVIF	Df	GVIF^(1/(2*Df))
ExerciseGuidelines	1.0657	1	1.0323
Sex	1.2176	1	1.1034
AlcoholRisk	1.0653	1	1.0321
SmokingDuration	1.2320	6	1.0175
NonSchoolQuals	1.1548	1	1.0746
IncomeDecile	1.5243	9	1.0237
Age	1.3539	1	1.1636
Waist	1.1807	1	1.0866

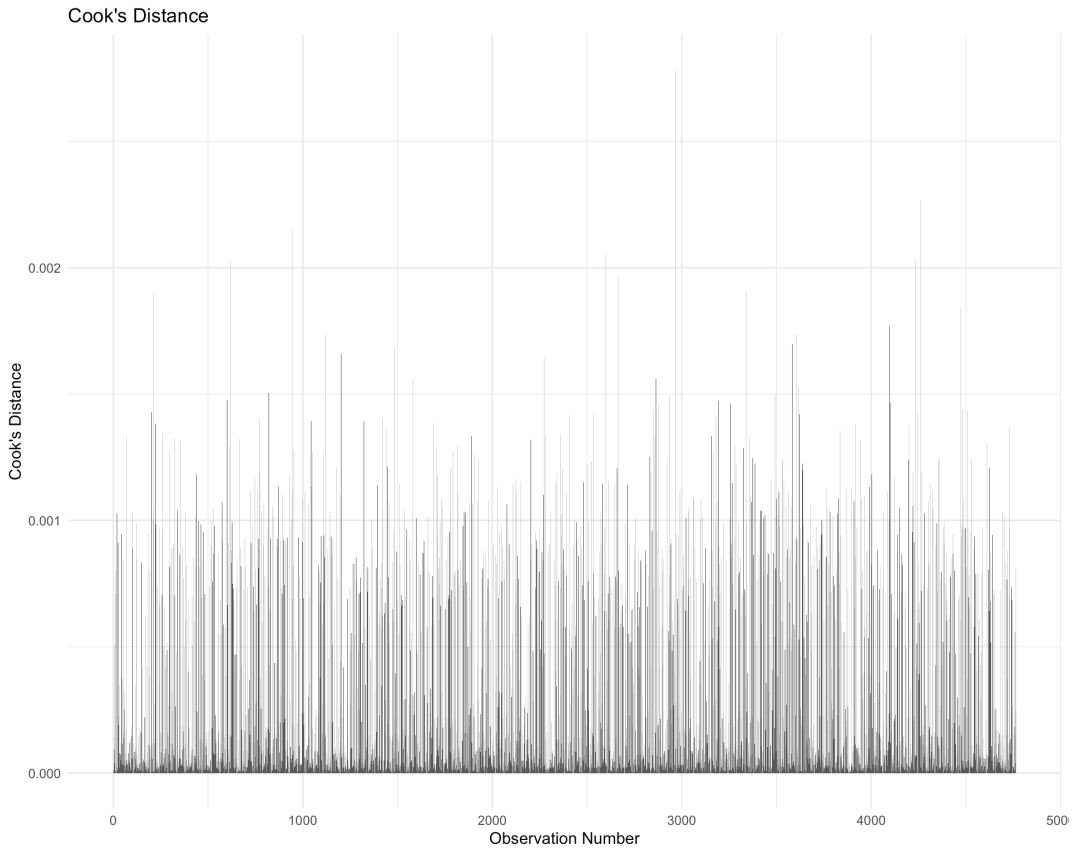
▼ Code

```
# Influential Observations
ggplot(ALT.logistic) + stat_cooks_obs() + theme_minimal()
```



▼ Code

```
ggplot(GGT.logistic) + stat_cooks_obs() + theme_minimal()
```



▼ Code

```
# Also fit GAM as well, to check linearity of logit assumptions
ALT.gam = gam(ALT2 ~ ExerciseGuidelines + Sex + AlcoholRisk +
    SmokingDuration + NonSchoolQuals + IncomeDecile + s(Age) + s
    data = dat.full, family = "binomial")
GGT.gam = gam(GGT2 ~ ExerciseGuidelines + Sex + AlcoholRisk +
    SmokingDuration + NonSchoolQuals + IncomeDecile + s(Age) + s
    data = dat.full, family = "binomial")
tidy(ALT.gam) %>% kable(caption = "GAM Spline Variables for ALT"
                        digits = 4)
```

GAM Spline Variables for ALT

term	edf	ref.df	statistic	p.value
s(Age)	5.7018	6.8430	85.1330	0
s(Waist)	3.7044	4.6282	242.7697	0

▼ Code

```
tidy(GGT.gam) %>% kable(caption = "GAM Spline Variables for GGT"
                        digits = 4)
```

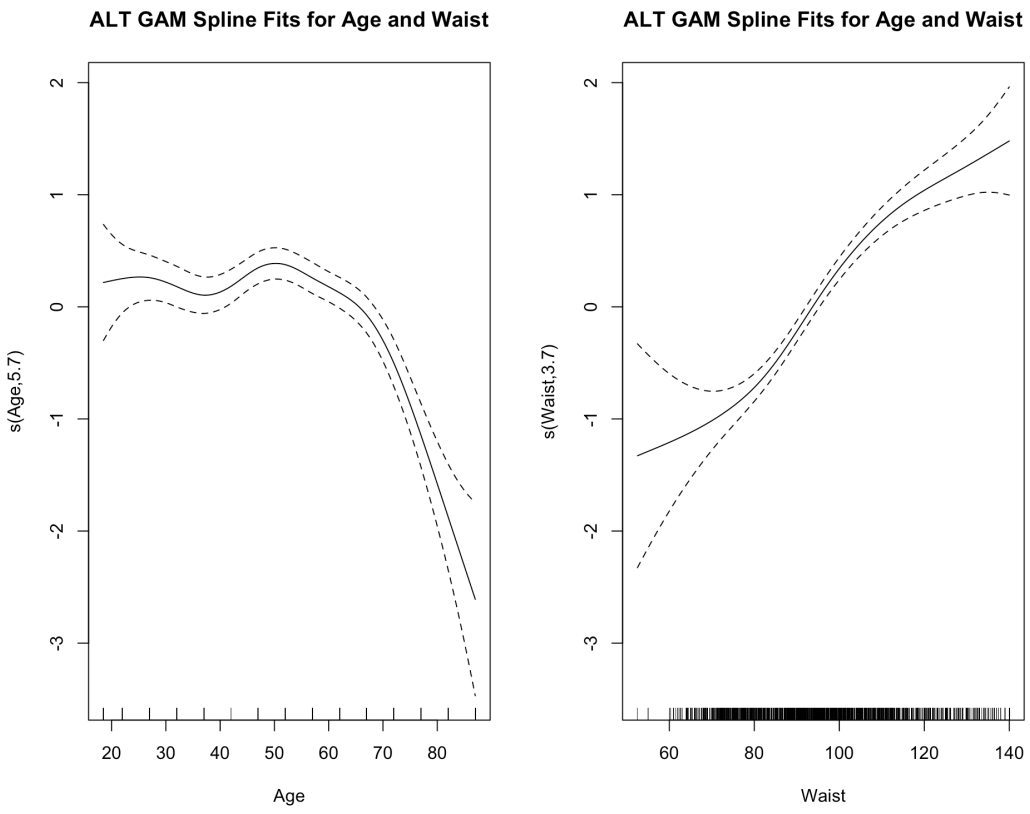
GAM Spline Variables for GGT

term	edf	ref.df	statistic	p.value
s(Age)	2.6942	3.4033	26.1092	0
s(Waist)	1.2670	1.4910	152.4728	0

We check using the `edf` value where `edf>1` suggests that there is a departure from linearity, so these models may not be appropriate. We can visualise the curves below as well.

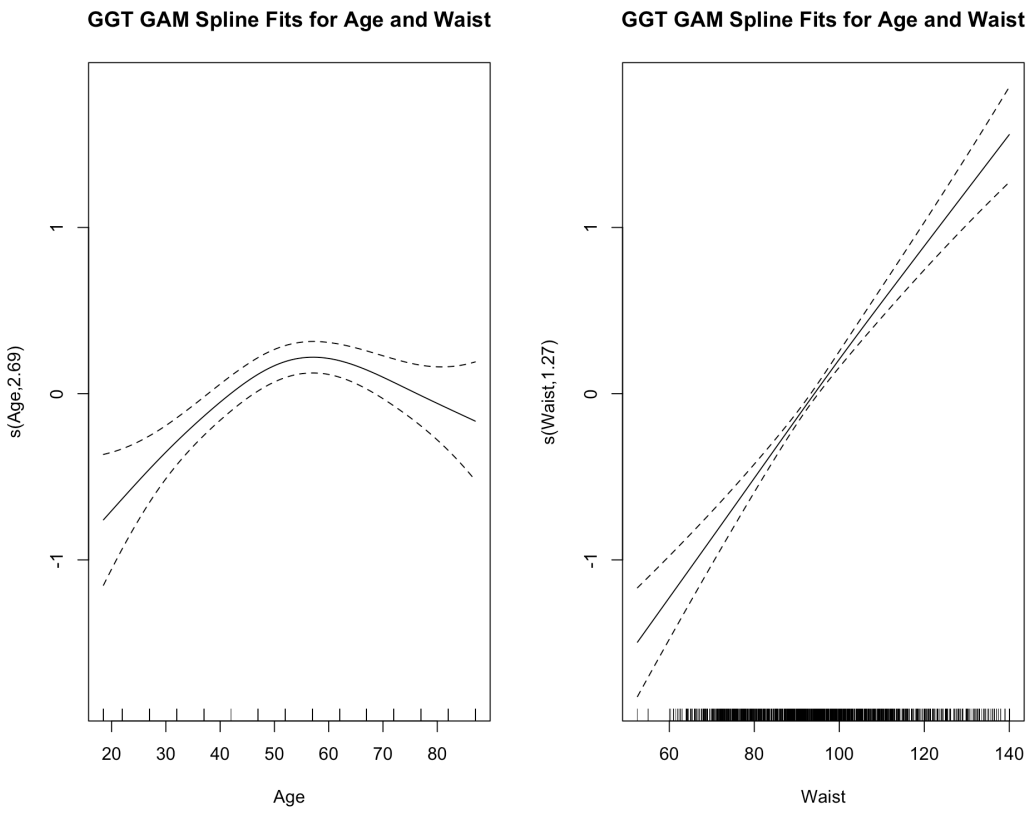
▼ Code

```
# GAM plots
par(mfrow = c(1, 2))
plot(ALT.gam, main = "ALT GAM Spline Fits for Age and Waist")
```



▼ Code

```
par(mfrow = c(1, 2))
plot(GGT.gam, main = "GGT GAM Spline Fits for Age and Waist")
```



This suggests that perhaps the binary logistic regression does not satisfy the assumptions as well as we had hoped. I think we can still sort of use it as a benchmark against our main multinomial logistic regression model, but be careful when analysing it's results.

Ordinal Logistic Regression

Another thing we can try once more is ordinal logistic regression, at least to check if the proportional odds assumption is satisfied in this new data set.

▼ Code

```
# Fit ordinal models
ALT.ordinal = polr(formula = ALT3 ~ ExerciseGuidelines + Sex + A
    SmokingDuration + NonSchoolQuals + IncomeDecile + Age + Wais
    data = dat.full, Hess = TRUE)

GGT.ordinal = polr(formula = GGT3 ~ ExerciseGuidelines + Sex + A
    SmokingDuration + NonSchoolQuals + IncomeDecile + Age + Wais
    data = dat.full, Hess = TRUE)

tidy(ALT.ordinal) %>% kable(caption = "ALT Ordinal Logistic Regr
                            digits = 4, align = "ccccc")
```

ALT Ordinal Logistic Regression Summary

term	estimate	std.error	statistic	coef.type
ExerciseGuidelinesNo	0.0166	0.0734	0.2254	coefficient
SexMale	0.3662	0.0775	4.7228	coefficient
AlcoholRiskMedium/High risk	0.2732	0.1031	2.6510	coefficient
SmokingDurationLess than 10 years	0.1696	0.1152	1.4724	coefficient
SmokingDuration10 to 19 years	0.2047	0.1118	1.8314	coefficient
SmokingDuration20 to 29 years	0.1167	0.1209	0.9653	coefficient
SmokingDuration30 to 39 years	0.2340	0.1390	1.6833	coefficient
SmokingDuration40 to 49 years	0.2861	0.1664	1.7194	coefficient
SmokingDuration50 years and over	-0.5367	0.3314	-1.6195	coefficient
NonSchoolQualsYes	0.1531	0.0800	1.9136	coefficient
IncomeDecile2	-0.4376	0.1874	-2.3354	coefficient
IncomeDecile3	-0.4111	0.1878	-2.1887	coefficient
IncomeDecile4	-0.2148	0.1795	-1.1964	coefficient
IncomeDecile5	-0.1700	0.1835	-0.9263	coefficient
IncomeDecile6	-0.1599	0.1823	-0.8772	coefficient
IncomeDecile7	-0.3438	0.1818	-1.8913	coefficient
IncomeDecile8	-0.1138	0.1782	-0.6385	coefficient
IncomeDecile9	-0.1528	0.1748	-0.8740	coefficient
IncomeDecile10	-0.2502	0.1789	-1.3983	coefficient
Age	-0.0160	0.0026	-6.1461	coefficient
Waist	0.0434	0.0026	16.5596	coefficient
Low|Borderline	4.7626	0.2924	16.2861	scale
Borderline|High	5.6117	0.2963	18.9408	scale

▼ Code

```
tidy(GGT.ordinal) %>% kable(caption = "GGT Ordinal Logistic Regr
                            digits = 4, align = "ccccc")
```

GGT Ordinal Logistic Regression Summary

term	estimate	std.error	statistic	coef.type

ExerciseGuidelinesNo	0.2539	0.0797	3.1858	coefficient
SexMale	-0.4318	0.0850	-5.0805	coefficient
AlcoholRiskMedium/High risk	0.8273	0.1020	8.1092	coefficient
SmokingDurationLess than 10 years	0.2163	0.1311	1.6506	coefficient
SmokingDuration10 to 19 years	0.2726	0.1236	2.2053	coefficient
SmokingDuration20 to 29 years	0.4013	0.1250	3.2114	coefficient
SmokingDuration30 to 39 years	0.3813	0.1407	2.7108	coefficient
SmokingDuration40 to 49 years	0.5120	0.1620	3.1597	coefficient
SmokingDuration50 years and over	0.5763	0.2440	2.3618	coefficient
NonSchoolQualsYes	-0.0740	0.0842	-0.8780	coefficient
IncomeDecile2	-0.3442	0.1882	-1.8286	coefficient
IncomeDecile3	-0.4652	0.1860	-2.5017	coefficient
IncomeDecile4	-0.3460	0.1840	-1.8801	coefficient
IncomeDecile5	-0.4281	0.1931	-2.2177	coefficient
IncomeDecile6	-0.3969	0.1966	-2.0190	coefficient
IncomeDecile7	-0.5384	0.1961	-2.7461	coefficient
IncomeDecile8	-0.2458	0.1913	-1.2847	coefficient
IncomeDecile9	-0.4817	0.1941	-2.4810	coefficient
IncomeDecile10	-0.2849	0.1964	-1.4502	coefficient
Age	0.0088	0.0028	3.0963	coefficient
Waist	0.0366	0.0028	13.2288	coefficient
Low|Borderline	5.3023	0.3248	16.3244	scale
Borderline|High	5.7599	0.3267	17.6300	scale

We see that the test statistic for `ExerciseGuidelines` under ALT is small while under GGT is large, again agreeing with our previous results on significance. Next, we use the Brant test and check AICs as well to see how much “better” the ordinal models would have been regardless of whether they satisfy their main assumption or not.

▼ Code

```
# Brant test
a1 = brant(ALT.ordinal)["Omnibus", ]
a2 = brant(GGT.ordinal)["Omnibus", ]
res = rbind(a1, a2)
rownames(res) = c("ALT", "GGT")
res %>% kable(caption = "Brant Omnibus test for ALT and GGT",
```

```
                align = "ccc")
```

▼ Code

```
# AIC tables
AIC(ALT.multinom, ALT.ordinal) %>%
  kable(caption = "AIC of ALT Models", align = "cc")
```

AIC of ALT Models

	df	AIC
ALT.multinom	44	6253.172
ALT.ordinal	23	6272.136

▼ Code

```
AIC(GGT.multinom, GGT.ordinal) %>%
  kable(caption = "AIC of GGT Models", align = "cc")
```

AIC of GGT Models

	df	AIC
GGT.multinom	44	5357.065
GGT.ordinal	23	5350.087

▼ Code

```
# BIC tables as well
BIC(ALT.multinom, ALT.ordinal) %>%
  kable(caption = "BIC of ALT Models", align = "cc")
```

BIC of ALT Models

	df	BIC
ALT.multinom	44	6537.782
ALT.ordinal	23	6420.910

▼ Code

```
BIC(GGT.multinom, GGT.ordinal) %>%
  kable(caption = "BIC of GGT Models", align = "cc")
```

BIC of GGT Models

BIC of GGT Models

	df	BIC
GGT.multinom	44	5641.676
GGT.ordinal	23	5498.861

The Brant test tells us that the POA is still violated, so we should not use this model. Hence, the main model remains the multinomial logistic regression model, but the benefit of all the previous analysis is that the model results are quite robust in that `ExerciseGuidelines` does not have a significant association with ALT, but it does with GGT across all our different models.

Summary

The bottom line for the rework is that physical activity showed little evidence of an adjusted association with ALT, but consistently showed an association with GGT, suggesting that the two liver biomarkers may capture different aspects of the relationship between lifestyle behaviours and liver health.

My next step would be to remake the causal inference DAGs to add all the new confounders, and I think that may be enough for now. I may do decision trees and random forests as well, since we have clearly seen a non-linear relationship for ALT and GGT.